FFNN Final Report

August 18, 2025

[1]:
```python
# Import necessary libraries
import pandas as pd
import numpy as np
from scipy.sparse import csr_matrix, hstack
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.layers import LeakyReLU
from tensorflow.keras.layers import ReLU
from tensorflow.keras.optimizers import Adam, RMSprop, SGD
from tensorflow.keras.callbacks import EarlyStopping
import matplotlib.pyplot as plt
import pyhere
```

[2]:
```python
train_path = pyhere.here('Data', 'train_data.xlsx')
test_path = pyhere.here('Data', 'test_data.xlsx')
train_data = pd.read_excel(train_path)
test_data = pd.read_excel(test_path)
# Use dropna() to remove any rows with missing values
train_data = train_data.dropna()
test_data = test_data.dropna()
print(train_data.head())
```

```
           CountryName CountryCode  year  HDI_Index  \
0   Antigua and Barbuda         ATG  2022      0.848
9   Antigua and Barbuda         ATG  2008      0.830
10  Antigua and Barbuda         ATG  2009      0.827
11  Antigua and Barbuda         ATG  2010      0.828
12  Antigua and Barbuda         ATG  2011      0.828

                  HDI_Category  FoodIndex  ElectricAccess  PopDensity  \
0   Very high human development      89.95           100.0  211.000000
9   Very high human development     145.98           100.0  188.436364
10  Very high human development     141.40            98.5  191.302273
11  Very high human development     102.49            98.7  193.909091
12  Very high human development     104.99            94.6  196.209091
```

```
    PopInSlums  WaterStress  …  IncomeGroup  YearlyChgInternet  \
0     2.646340     8.461538  …  High income           0.997844
9     2.646317     9.230769  …  High income           3.164091
10    2.646311     9.038462  …  High income           3.164091
11    2.646300     8.846154  …  High income           3.831893
12    2.646300     8.653846  …  High income           3.831893

    YearlyChgHDI  Lag1_InternetUsersPct  Lag2_InternetUsersPct  \
0          0.005               8.247411               7.147796
9          0.003               5.957120               5.079504
10        -0.003               6.242918               5.336653
11         0.001               6.508026               5.571669
12         0.000               6.815164               5.788459

    Lag1_YearlyChgInternet  Lag2_YearlyChgInternet  \
0                 0.993070                0.908060
9                 3.128215                2.432146
10                3.128215                3.116179
11                3.128215                3.116179
12                3.782704                3.116179

    Cumulative3yrChg_InternetUsersPct  GovernanceScore  HealthSpend
0                                 3.2         0.180504   853.834335
9                                11.0         0.904398   523.176632
10                               12.0         0.929469   472.044125
11                               13.0         0.914123   573.175838
12                               14.0         0.898586   562.880747

[5 rows x 30 columns]
```

[3]:
```python
train_target = train_data['InternetUsersPct']
test_target = test_data['InternetUsersPct']

train_predictors = train_data.drop('InternetUsersPct', axis=1)
test_predictors = test_data.drop('InternetUsersPct', axis=1)
```

[4]:
```python
train_target = train_data['InternetUsersPct']
test_target = test_data['InternetUsersPct']

train_predictors = train_data.drop('InternetUsersPct', axis=1)
test_predictors = test_data.drop('InternetUsersPct', axis=1)

# Combine the predictor data frames to ensure consistent encoding
combined_predictors = pd.concat([train_predictors, test_predictors],␣
 ↪ignore_index=True)
```

```python
# Identify numerical and categorical columns
numerical_cols = combined_predictors.select_dtypes(include=np.number).columns
categorical_cols = combined_predictors.select_dtypes(include='object').columns

# Initialize MinMaxScaler
scaler = MinMaxScaler()

# Fit the scaler on the combined numerical data to learn the min/max values
combined_predictors[numerical_cols] = scaler.
 ↪fit_transform(combined_predictors[numerical_cols])

# Initialize OneHotEncoder to create sparse output and handle unknown categories
ohe = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

# Fit the encoder on the combined data to learn all possible categories
combined_categorical_encoded = ohe.
 ↪fit_transform(combined_predictors[categorical_cols])

# Combine the encoded categorical data with the numerical data
combined_features = np.concatenate([combined_predictors[numerical_cols].values,
                                    combined_categorical_encoded], axis=1)

# Split the combined feature matrix back into training and test sets
train_features = combined_features[:len(train_predictors)]
test_features = combined_features[len(train_predictors):]

# Features split
train_features, test_features, train_target, test_target = train_test_split(
    combined_features, train_target, test_size=0.2, random_state=42
)
```

[5]:
```python
print("\n--- Training Model with Tanh and RMSProp---")
model_tanh_rms = Sequential()
model_tanh_rms.add(Dense(256, input_dim=train_features.shape[1],␣
 ↪activation='tanh'))
model_tanh_rms.add(Dense(128, activation='tanh'))
model_tanh_rms.add(Dense(64, activation='tanh'))
model_tanh_rms.add(Dense(1, activation='linear'))

model_tanh_rms.compile(optimizer=RMSprop(learning_rate=0.001), loss="mse",␣
 ↪metrics=["mae"])

# Define early stopping
early_stop_tanh = EarlyStopping(monitor='val_loss', patience=50, verbose=1,␣
 ↪restore_best_weights=True)

model_tanh_rms.fit(train_features, train_target,
```

```python
                    epochs=1000,
                    batch_size=32,
                    validation_data=(test_features, test_target),
                    callbacks=[early_stop_tanh],
                    verbose=1)

loss_tanh_rms =model_tanh_rms.evaluate(test_features, test_target, verbose=0)

history_tanh = model_tanh_rms.fit(train_features, train_target,
                    epochs=1000,
                    batch_size=32,
                    validation_split=0.2,
                    callbacks=[early_stop_tanh],
                    verbose=1)

tanh_results = model_tanh_rms.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", tanh_results)

# Plot the training and validation MAE
mae_tanh = history_tanh.history['mae']
val_mae_tanh = history_tanh.history['val_mae']
epochs = range(1, len(mae_tanh) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_tanh, 'b', label='Training MAE')
plt.plot(epochs, val_mae_tanh, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Tanh + RMSprop)')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()
```

```
--- Training Model with Tanh and RMSProp---
Epoch 1/1000
```

```
C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
57/57            1s 8ms/step - loss:
3.1446 - mae: 1.2538 - val_loss: 1.0091 - val_mae: 0.8191
Epoch 2/1000
57/57            0s 4ms/step - loss:
0.6949 - mae: 0.6644 - val_loss: 0.8733 - val_mae: 0.8129
Epoch 3/1000
57/57            0s 4ms/step - loss:
0.4548 - mae: 0.5162 - val_loss: 0.4063 - val_mae: 0.5321
```

```
Epoch 4/1000
57/57            0s 4ms/step - loss:
0.3630 - mae: 0.4485 - val_loss: 0.2582 - val_mae: 0.4057
Epoch 5/1000
57/57            0s 4ms/step - loss:
0.2864 - mae: 0.4154 - val_loss: 0.2496 - val_mae: 0.4084
Epoch 6/1000
57/57            0s 4ms/step - loss:
0.3157 - mae: 0.4487 - val_loss: 0.1516 - val_mae: 0.3026
Epoch 7/1000
57/57            0s 4ms/step - loss:
0.2294 - mae: 0.3682 - val_loss: 0.1088 - val_mae: 0.2524
Epoch 8/1000
57/57            0s 4ms/step - loss:
0.2511 - mae: 0.3995 - val_loss: 0.1233 - val_mae: 0.2701
Epoch 9/1000
57/57            0s 4ms/step - loss:
0.2096 - mae: 0.3585 - val_loss: 0.1698 - val_mae: 0.3313
Epoch 10/1000
57/57            0s 4ms/step - loss:
0.1840 - mae: 0.3358 - val_loss: 0.4067 - val_mae: 0.5425
Epoch 11/1000
57/57            0s 4ms/step - loss:
0.1920 - mae: 0.3366 - val_loss: 0.0763 - val_mae: 0.2095
Epoch 12/1000
57/57            0s 4ms/step - loss:
0.1603 - mae: 0.3062 - val_loss: 0.2248 - val_mae: 0.3928
Epoch 13/1000
57/57            0s 4ms/step - loss:
0.1674 - mae: 0.3228 - val_loss: 0.3606 - val_mae: 0.5092
Epoch 14/1000
57/57            0s 4ms/step - loss:
0.1497 - mae: 0.3062 - val_loss: 0.1209 - val_mae: 0.2652
Epoch 15/1000
57/57            0s 4ms/step - loss:
0.1475 - mae: 0.2971 - val_loss: 0.1233 - val_mae: 0.2709
Epoch 16/1000
57/57            0s 4ms/step - loss:
0.1359 - mae: 0.2873 - val_loss: 0.1722 - val_mae: 0.3382
Epoch 17/1000
57/57            0s 4ms/step - loss:
0.1264 - mae: 0.2843 - val_loss: 0.0701 - val_mae: 0.2024
Epoch 18/1000
57/57            0s 4ms/step - loss:
0.1247 - mae: 0.2645 - val_loss: 0.1214 - val_mae: 0.2821
Epoch 19/1000
57/57            0s 4ms/step - loss:
0.1194 - mae: 0.2650 - val_loss: 0.1580 - val_mae: 0.3290
```

```
Epoch 20/1000
57/57            0s 4ms/step - loss:
0.1139 - mae: 0.2654 - val_loss: 0.0788 - val_mae: 0.2134
Epoch 21/1000
57/57            0s 4ms/step - loss:
0.1159 - mae: 0.2626 - val_loss: 0.0841 - val_mae: 0.2243
Epoch 22/1000
57/57            0s 4ms/step - loss:
0.1110 - mae: 0.2546 - val_loss: 0.0623 - val_mae: 0.1858
Epoch 23/1000
57/57            0s 4ms/step - loss:
0.0973 - mae: 0.2344 - val_loss: 0.0564 - val_mae: 0.1815
Epoch 24/1000
57/57            0s 4ms/step - loss:
0.1124 - mae: 0.2569 - val_loss: 0.0541 - val_mae: 0.1721
Epoch 25/1000
57/57            0s 4ms/step - loss:
0.0933 - mae: 0.2345 - val_loss: 0.1076 - val_mae: 0.2649
Epoch 26/1000
57/57            0s 4ms/step - loss:
0.0998 - mae: 0.2429 - val_loss: 0.0917 - val_mae: 0.2365
Epoch 27/1000
57/57            0s 4ms/step - loss:
0.0903 - mae: 0.2356 - val_loss: 0.1190 - val_mae: 0.2732
Epoch 28/1000
57/57            0s 4ms/step - loss:
0.0901 - mae: 0.2333 - val_loss: 0.1128 - val_mae: 0.2561
Epoch 29/1000
57/57            0s 4ms/step - loss:
0.0923 - mae: 0.2346 - val_loss: 0.0912 - val_mae: 0.2337
Epoch 30/1000
57/57            0s 4ms/step - loss:
0.0872 - mae: 0.2302 - val_loss: 0.0523 - val_mae: 0.1664
Epoch 31/1000
57/57            0s 4ms/step - loss:
0.0833 - mae: 0.2171 - val_loss: 0.0990 - val_mae: 0.2403
Epoch 32/1000
57/57            0s 4ms/step - loss:
0.0829 - mae: 0.2187 - val_loss: 0.0762 - val_mae: 0.2083
Epoch 33/1000
57/57            0s 4ms/step - loss:
0.0730 - mae: 0.2081 - val_loss: 0.0715 - val_mae: 0.2017
Epoch 34/1000
57/57            0s 4ms/step - loss:
0.0858 - mae: 0.2215 - val_loss: 0.0618 - val_mae: 0.1889
Epoch 35/1000
57/57            0s 4ms/step - loss:
0.0720 - mae: 0.2044 - val_loss: 0.0501 - val_mae: 0.1667
```

```
Epoch 36/1000
57/57            0s 4ms/step - loss:
0.0766 - mae: 0.2117 - val_loss: 0.0648 - val_mae: 0.1896
Epoch 37/1000
57/57            0s 4ms/step - loss:
0.0751 - mae: 0.2113 - val_loss: 0.0606 - val_mae: 0.1903
Epoch 38/1000
57/57            0s 4ms/step - loss:
0.0707 - mae: 0.2016 - val_loss: 0.1310 - val_mae: 0.2933
Epoch 39/1000
57/57            0s 4ms/step - loss:
0.0722 - mae: 0.2077 - val_loss: 0.0583 - val_mae: 0.1837
Epoch 40/1000
57/57            0s 4ms/step - loss:
0.0700 - mae: 0.2048 - val_loss: 0.0816 - val_mae: 0.2256
Epoch 41/1000
57/57            0s 4ms/step - loss:
0.0646 - mae: 0.1891 - val_loss: 0.1281 - val_mae: 0.2906
Epoch 42/1000
57/57            0s 4ms/step - loss:
0.0676 - mae: 0.1956 - val_loss: 0.0436 - val_mae: 0.1548
Epoch 43/1000
57/57            0s 4ms/step - loss:
0.0685 - mae: 0.2039 - val_loss: 0.0516 - val_mae: 0.1764
Epoch 44/1000
57/57            0s 4ms/step - loss:
0.0673 - mae: 0.1985 - val_loss: 0.0471 - val_mae: 0.1607
Epoch 45/1000
57/57            0s 4ms/step - loss:
0.0678 - mae: 0.1968 - val_loss: 0.1074 - val_mae: 0.2628
Epoch 46/1000
57/57            0s 4ms/step - loss:
0.0605 - mae: 0.1879 - val_loss: 0.0782 - val_mae: 0.2119
Epoch 47/1000
57/57            0s 4ms/step - loss:
0.0575 - mae: 0.1867 - val_loss: 0.1310 - val_mae: 0.2961
Epoch 48/1000
57/57            0s 4ms/step - loss:
0.0600 - mae: 0.1826 - val_loss: 0.0433 - val_mae: 0.1552
Epoch 49/1000
57/57            0s 4ms/step - loss:
0.0606 - mae: 0.1888 - val_loss: 0.0373 - val_mae: 0.1435
Epoch 50/1000
57/57            0s 4ms/step - loss:
0.0596 - mae: 0.1900 - val_loss: 0.0488 - val_mae: 0.1656
Epoch 51/1000
57/57            0s 4ms/step - loss:
0.0574 - mae: 0.1854 - val_loss: 0.0649 - val_mae: 0.1994
```

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Epoch 52/1000
57/57            0s 4ms/step - loss:
0.0579 - mae: 0.1840 - val_loss: 0.1365 - val_mae: 0.3038
Epoch 53/1000
57/57            0s 4ms/step - loss:
0.0541 - mae: 0.1768 - val_loss: 0.0556 - val_mae: 0.1799
Epoch 54/1000
57/57            0s 4ms/step - loss:
0.0553 - mae: 0.1807 - val_loss: 0.0411 - val_mae: 0.1555
Epoch 55/1000
57/57            0s 4ms/step - loss:
0.0560 - mae: 0.1794 - val_loss: 0.0498 - val_mae: 0.1753
Epoch 56/1000
57/57            0s 4ms/step - loss:
0.0526 - mae: 0.1764 - val_loss: 0.0580 - val_mae: 0.1856
Epoch 57/1000
57/57            0s 4ms/step - loss:
0.0493 - mae: 0.1685 - val_loss: 0.0744 - val_mae: 0.2117
Epoch 58/1000
57/57            0s 4ms/step - loss:
0.0506 - mae: 0.1731 - val_loss: 0.1292 - val_mae: 0.2909
Epoch 59/1000
57/57            0s 4ms/step - loss:
0.0517 - mae: 0.1682 - val_loss: 0.0788 - val_mae: 0.2206
Epoch 60/1000
57/57            0s 4ms/step - loss:
0.0501 - mae: 0.1699 - val_loss: 0.0394 - val_mae: 0.1498
Epoch 61/1000
57/57            0s 4ms/step - loss:
0.0445 - mae: 0.1576 - val_loss: 0.1659 - val_mae: 0.3318
Epoch 62/1000
57/57            0s 4ms/step - loss:
0.0519 - mae: 0.1723 - val_loss: 0.0836 - val_mae: 0.2231
Epoch 63/1000
57/57            0s 4ms/step - loss:
0.0444 - mae: 0.1606 - val_loss: 0.0350 - val_mae: 0.1410
Epoch 64/1000
57/57            0s 4ms/step - loss:
0.0500 - mae: 0.1698 - val_loss: 0.0382 - val_mae: 0.1509
Epoch 65/1000
57/57            0s 4ms/step - loss:
0.0487 - mae: 0.1680 - val_loss: 0.0578 - val_mae: 0.1847
Epoch 66/1000
57/57            0s 4ms/step - loss:
0.0462 - mae: 0.1627 - val_loss: 0.0361 - val_mae: 0.1453
Epoch 67/1000
57/57            0s 4ms/step - loss:
0.0440 - mae: 0.1588 - val_loss: 0.0405 - val_mae: 0.1450
```

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Epoch 68/1000
57/57            0s 4ms/step - loss:
0.0479 - mae: 0.1632 - val_loss: 0.0621 - val_mae: 0.1842
Epoch 69/1000
57/57            0s 4ms/step - loss:
0.0420 - mae: 0.1532 - val_loss: 0.1489 - val_mae: 0.3130
Epoch 70/1000
57/57            0s 4ms/step - loss:
0.0445 - mae: 0.1629 - val_loss: 0.0623 - val_mae: 0.1991
Epoch 71/1000
57/57            0s 4ms/step - loss:
0.0409 - mae: 0.1542 - val_loss: 0.0369 - val_mae: 0.1456
Epoch 72/1000
57/57            0s 4ms/step - loss:
0.0442 - mae: 0.1586 - val_loss: 0.0395 - val_mae: 0.1488
Epoch 73/1000
57/57            0s 4ms/step - loss:
0.0404 - mae: 0.1491 - val_loss: 0.0374 - val_mae: 0.1442
Epoch 74/1000
57/57            0s 4ms/step - loss:
0.0398 - mae: 0.1528 - val_loss: 0.0486 - val_mae: 0.1597
Epoch 75/1000
57/57            0s 4ms/step - loss:
0.0411 - mae: 0.1541 - val_loss: 0.0370 - val_mae: 0.1408
Epoch 76/1000
57/57            0s 4ms/step - loss:
0.0427 - mae: 0.1559 - val_loss: 0.0571 - val_mae: 0.1842
Epoch 77/1000
57/57            0s 4ms/step - loss:
0.0367 - mae: 0.1459 - val_loss: 0.0428 - val_mae: 0.1584
Epoch 78/1000
57/57            0s 4ms/step - loss:
0.0391 - mae: 0.1532 - val_loss: 0.0620 - val_mae: 0.1957
Epoch 79/1000
57/57            0s 4ms/step - loss:
0.0393 - mae: 0.1557 - val_loss: 0.0371 - val_mae: 0.1422
Epoch 80/1000
57/57            0s 4ms/step - loss:
0.0389 - mae: 0.1491 - val_loss: 0.0375 - val_mae: 0.1448
Epoch 81/1000
57/57            0s 4ms/step - loss:
0.0389 - mae: 0.1444 - val_loss: 0.0640 - val_mae: 0.1945
Epoch 82/1000
57/57            0s 4ms/step - loss:
0.0372 - mae: 0.1468 - val_loss: 0.0450 - val_mae: 0.1605
Epoch 83/1000
57/57            0s 4ms/step - loss:
0.0377 - mae: 0.1490 - val_loss: 0.0440 - val_mae: 0.1570
```

```
Epoch 84/1000
57/57            0s 4ms/step - loss:
0.0370 - mae: 0.1473 - val_loss: 0.0745 - val_mae: 0.2196
Epoch 85/1000
57/57            0s 4ms/step - loss:
0.0379 - mae: 0.1486 - val_loss: 0.0396 - val_mae: 0.1527
Epoch 86/1000
57/57            0s 4ms/step - loss:
0.0356 - mae: 0.1442 - val_loss: 0.0272 - val_mae: 0.1240
Epoch 87/1000
57/57            0s 4ms/step - loss:
0.0333 - mae: 0.1390 - val_loss: 0.0728 - val_mae: 0.2118
Epoch 88/1000
57/57            0s 4ms/step - loss:
0.0359 - mae: 0.1472 - val_loss: 0.0336 - val_mae: 0.1333
Epoch 89/1000
57/57            0s 4ms/step - loss:
0.0355 - mae: 0.1412 - val_loss: 0.1101 - val_mae: 0.2791
Epoch 90/1000
57/57            0s 4ms/step - loss:
0.0357 - mae: 0.1420 - val_loss: 0.0388 - val_mae: 0.1473
Epoch 91/1000
57/57            0s 4ms/step - loss:
0.0337 - mae: 0.1393 - val_loss: 0.0536 - val_mae: 0.1769
Epoch 92/1000
57/57            0s 4ms/step - loss:
0.0347 - mae: 0.1426 - val_loss: 0.0426 - val_mae: 0.1561
Epoch 93/1000
57/57            0s 4ms/step - loss:
0.0345 - mae: 0.1406 - val_loss: 0.0409 - val_mae: 0.1592
Epoch 94/1000
57/57            0s 4ms/step - loss:
0.0369 - mae: 0.1437 - val_loss: 0.0310 - val_mae: 0.1234
Epoch 95/1000
57/57            0s 4ms/step - loss:
0.0333 - mae: 0.1370 - val_loss: 0.0756 - val_mae: 0.2080
Epoch 96/1000
57/57            0s 4ms/step - loss:
0.0297 - mae: 0.1298 - val_loss: 0.0333 - val_mae: 0.1363
Epoch 97/1000
57/57            0s 4ms/step - loss:
0.0351 - mae: 0.1463 - val_loss: 0.0323 - val_mae: 0.1356
Epoch 98/1000
57/57            0s 4ms/step - loss:
0.0305 - mae: 0.1347 - val_loss: 0.0325 - val_mae: 0.1314
Epoch 99/1000
57/57            0s 4ms/step - loss:
0.0316 - mae: 0.1384 - val_loss: 0.0331 - val_mae: 0.1372
```

```
Epoch 100/1000
57/57            0s 4ms/step - loss:
0.0302 - mae: 0.1317 - val_loss: 0.0481 - val_mae: 0.1630
Epoch 101/1000
57/57            0s 4ms/step - loss:
0.0289 - mae: 0.1288 - val_loss: 0.0917 - val_mae: 0.2556
Epoch 102/1000
57/57            0s 4ms/step - loss:
0.0342 - mae: 0.1419 - val_loss: 0.0867 - val_mae: 0.2221
Epoch 103/1000
57/57            0s 4ms/step - loss:
0.0310 - mae: 0.1340 - val_loss: 0.0410 - val_mae: 0.1550
Epoch 104/1000
57/57            0s 4ms/step - loss:
0.0296 - mae: 0.1326 - val_loss: 0.0396 - val_mae: 0.1486
Epoch 105/1000
57/57            0s 4ms/step - loss:
0.0289 - mae: 0.1285 - val_loss: 0.0353 - val_mae: 0.1376
Epoch 106/1000
57/57            0s 4ms/step - loss:
0.0322 - mae: 0.1361 - val_loss: 0.0310 - val_mae: 0.1298
Epoch 107/1000
57/57            0s 4ms/step - loss:
0.0289 - mae: 0.1276 - val_loss: 0.0358 - val_mae: 0.1440
Epoch 108/1000
57/57            0s 4ms/step - loss:
0.0296 - mae: 0.1266 - val_loss: 0.0317 - val_mae: 0.1295
Epoch 109/1000
57/57            0s 4ms/step - loss:
0.0277 - mae: 0.1270 - val_loss: 0.0304 - val_mae: 0.1279
Epoch 110/1000
57/57            0s 4ms/step - loss:
0.0306 - mae: 0.1321 - val_loss: 0.0298 - val_mae: 0.1258
Epoch 111/1000
57/57            0s 4ms/step - loss:
0.0291 - mae: 0.1249 - val_loss: 0.0493 - val_mae: 0.1641
Epoch 112/1000
57/57            0s 4ms/step - loss:
0.0267 - mae: 0.1212 - val_loss: 0.0349 - val_mae: 0.1366
Epoch 113/1000
57/57            0s 4ms/step - loss:
0.0285 - mae: 0.1245 - val_loss: 0.0687 - val_mae: 0.2049
Epoch 114/1000
57/57            0s 4ms/step - loss:
0.0295 - mae: 0.1317 - val_loss: 0.0430 - val_mae: 0.1642
Epoch 115/1000
57/57            0s 4ms/step - loss:
0.0257 - mae: 0.1216 - val_loss: 0.0365 - val_mae: 0.1380
```

```
Epoch 116/1000
57/57            0s 4ms/step - loss:
0.0291 - mae: 0.1280 - val_loss: 0.0418 - val_mae: 0.1509
Epoch 117/1000
57/57            0s 4ms/step - loss:
0.0270 - mae: 0.1256 - val_loss: 0.0293 - val_mae: 0.1225
Epoch 118/1000
57/57            0s 4ms/step - loss:
0.0279 - mae: 0.1292 - val_loss: 0.0325 - val_mae: 0.1270
Epoch 119/1000
57/57            0s 4ms/step - loss:
0.0261 - mae: 0.1235 - val_loss: 0.0545 - val_mae: 0.1785
Epoch 120/1000
57/57            0s 4ms/step - loss:
0.0262 - mae: 0.1220 - val_loss: 0.0495 - val_mae: 0.1738
Epoch 121/1000
57/57            0s 4ms/step - loss:
0.0257 - mae: 0.1236 - val_loss: 0.0416 - val_mae: 0.1516
Epoch 122/1000
57/57            0s 4ms/step - loss:
0.0276 - mae: 0.1267 - val_loss: 0.0279 - val_mae: 0.1209
Epoch 123/1000
57/57            0s 4ms/step - loss:
0.0243 - mae: 0.1163 - val_loss: 0.0297 - val_mae: 0.1260
Epoch 124/1000
57/57            0s 4ms/step - loss:
0.0263 - mae: 0.1224 - val_loss: 0.0281 - val_mae: 0.1207
Epoch 125/1000
57/57            0s 4ms/step - loss:
0.0254 - mae: 0.1174 - val_loss: 0.0349 - val_mae: 0.1363
Epoch 126/1000
57/57            0s 4ms/step - loss:
0.0237 - mae: 0.1173 - val_loss: 0.0400 - val_mae: 0.1533
Epoch 127/1000
57/57            0s 4ms/step - loss:
0.0236 - mae: 0.1182 - val_loss: 0.0476 - val_mae: 0.1600
Epoch 128/1000
57/57            0s 4ms/step - loss:
0.0280 - mae: 0.1279 - val_loss: 0.0339 - val_mae: 0.1322
Epoch 129/1000
57/57            0s 4ms/step - loss:
0.0256 - mae: 0.1245 - val_loss: 0.0666 - val_mae: 0.1952
Epoch 130/1000
57/57            0s 4ms/step - loss:
0.0230 - mae: 0.1159 - val_loss: 0.0857 - val_mae: 0.2378
Epoch 131/1000
57/57            0s 4ms/step - loss:
0.0252 - mae: 0.1204 - val_loss: 0.0497 - val_mae: 0.1670
```

```
Epoch 132/1000
57/57              0s 4ms/step - loss:
0.0230 - mae: 0.1154 - val_loss: 0.0327 - val_mae: 0.1349
Epoch 133/1000
57/57              0s 4ms/step - loss:
0.0250 - mae: 0.1167 - val_loss: 0.0417 - val_mae: 0.1562
Epoch 134/1000
57/57              0s 4ms/step - loss:
0.0223 - mae: 0.1166 - val_loss: 0.0378 - val_mae: 0.1412
Epoch 135/1000
57/57              0s 4ms/step - loss:
0.0221 - mae: 0.1157 - val_loss: 0.0367 - val_mae: 0.1356
Epoch 136/1000
57/57              0s 4ms/step - loss:
0.0260 - mae: 0.1205 - val_loss: 0.0248 - val_mae: 0.1108
Epoch 137/1000
57/57              0s 4ms/step - loss:
0.0232 - mae: 0.1185 - val_loss: 0.0715 - val_mae: 0.2046
Epoch 138/1000
57/57              0s 4ms/step - loss:
0.0219 - mae: 0.1135 - val_loss: 0.0941 - val_mae: 0.2526
Epoch 139/1000
57/57              0s 4ms/step - loss:
0.0230 - mae: 0.1161 - val_loss: 0.0645 - val_mae: 0.2012
Epoch 140/1000
57/57              0s 4ms/step - loss:
0.0219 - mae: 0.1130 - val_loss: 0.0309 - val_mae: 0.1271
Epoch 141/1000
57/57              0s 4ms/step - loss:
0.0239 - mae: 0.1169 - val_loss: 0.0354 - val_mae: 0.1411
Epoch 142/1000
57/57              0s 4ms/step - loss:
0.0210 - mae: 0.1111 - val_loss: 0.0305 - val_mae: 0.1264
Epoch 143/1000
57/57              0s 4ms/step - loss:
0.0211 - mae: 0.1077 - val_loss: 0.0373 - val_mae: 0.1510
Epoch 144/1000
57/57              0s 4ms/step - loss:
0.0233 - mae: 0.1179 - val_loss: 0.0390 - val_mae: 0.1539
Epoch 145/1000
57/57              0s 4ms/step - loss:
0.0211 - mae: 0.1112 - val_loss: 0.0321 - val_mae: 0.1297
Epoch 146/1000
57/57              0s 4ms/step - loss:
0.0220 - mae: 0.1133 - val_loss: 0.0500 - val_mae: 0.1686
Epoch 147/1000
57/57              0s 4ms/step - loss:
0.0209 - mae: 0.1109 - val_loss: 0.0459 - val_mae: 0.1580
```

```
Epoch 148/1000
57/57            0s 4ms/step - loss:
0.0216 - mae: 0.1113 - val_loss: 0.0409 - val_mae: 0.1498
Epoch 149/1000
57/57            0s 4ms/step - loss:
0.0202 - mae: 0.1069 - val_loss: 0.0345 - val_mae: 0.1358
Epoch 150/1000
57/57            0s 4ms/step - loss:
0.0196 - mae: 0.1064 - val_loss: 0.0308 - val_mae: 0.1211
Epoch 151/1000
57/57            0s 4ms/step - loss:
0.0219 - mae: 0.1122 - val_loss: 0.0348 - val_mae: 0.1373
Epoch 152/1000
57/57            0s 4ms/step - loss:
0.0206 - mae: 0.1098 - val_loss: 0.0405 - val_mae: 0.1460
Epoch 153/1000
57/57            0s 4ms/step - loss:
0.0195 - mae: 0.1073 - val_loss: 0.0394 - val_mae: 0.1496
Epoch 154/1000
57/57            0s 4ms/step - loss:
0.0213 - mae: 0.1131 - val_loss: 0.0415 - val_mae: 0.1474
Epoch 155/1000
57/57            0s 4ms/step - loss:
0.0212 - mae: 0.1129 - val_loss: 0.0501 - val_mae: 0.1706
Epoch 156/1000
57/57            0s 4ms/step - loss:
0.0201 - mae: 0.1084 - val_loss: 0.0398 - val_mae: 0.1475
Epoch 157/1000
57/57            0s 4ms/step - loss:
0.0194 - mae: 0.1055 - val_loss: 0.0368 - val_mae: 0.1384
Epoch 158/1000
57/57            0s 4ms/step - loss:
0.0221 - mae: 0.1159 - val_loss: 0.0457 - val_mae: 0.1568
Epoch 159/1000
57/57            0s 4ms/step - loss:
0.0182 - mae: 0.1023 - val_loss: 0.0361 - val_mae: 0.1423
Epoch 160/1000
57/57            0s 4ms/step - loss:
0.0203 - mae: 0.1060 - val_loss: 0.0295 - val_mae: 0.1201
Epoch 161/1000
57/57            0s 4ms/step - loss:
0.0188 - mae: 0.1029 - val_loss: 0.0396 - val_mae: 0.1460
Epoch 162/1000
57/57            0s 4ms/step - loss:
0.0181 - mae: 0.1023 - val_loss: 0.0642 - val_mae: 0.2013
Epoch 163/1000
57/57            0s 4ms/step - loss:
0.0186 - mae: 0.1033 - val_loss: 0.0507 - val_mae: 0.1684
```

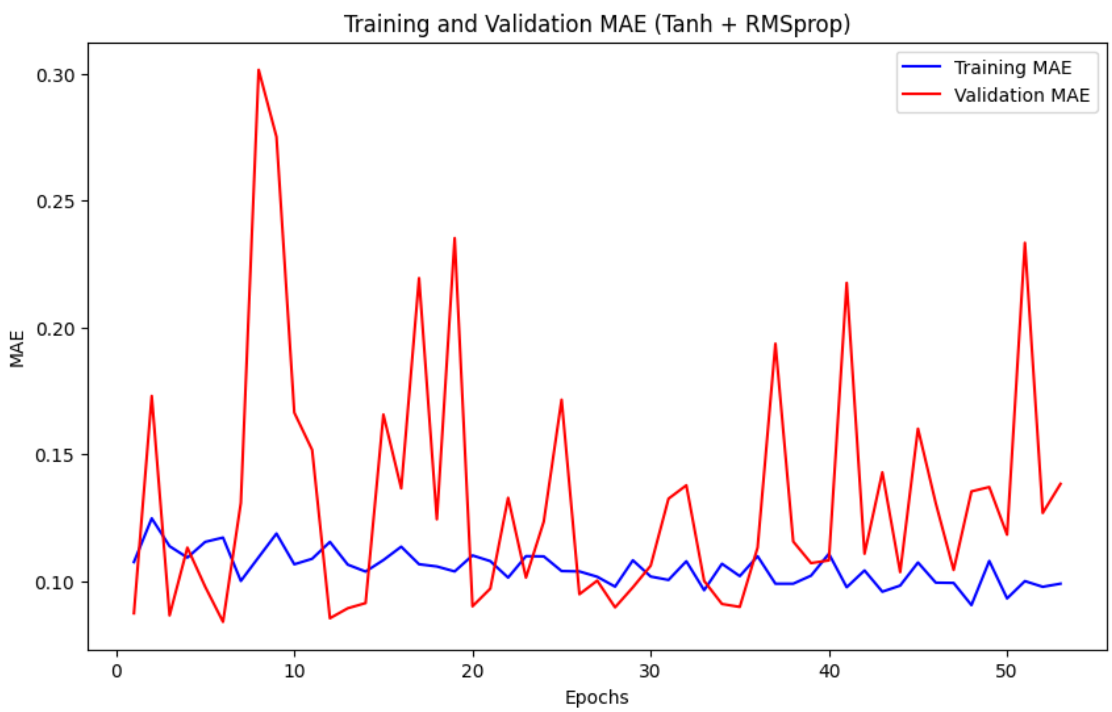
```
Epoch 164/1000
57/57            0s 4ms/step - loss:
0.0219 - mae: 0.1135 - val_loss: 0.0392 - val_mae: 0.1468
Epoch 165/1000
57/57            0s 4ms/step - loss:
0.0187 - mae: 0.1039 - val_loss: 0.0272 - val_mae: 0.1163
Epoch 166/1000
57/57            0s 4ms/step - loss:
0.0213 - mae: 0.1095 - val_loss: 0.0452 - val_mae: 0.1568
Epoch 167/1000
57/57            0s 4ms/step - loss:
0.0173 - mae: 0.1024 - val_loss: 0.0364 - val_mae: 0.1304
Epoch 168/1000
57/57            0s 4ms/step - loss:
0.0183 - mae: 0.1023 - val_loss: 0.0307 - val_mae: 0.1235
Epoch 169/1000
57/57            0s 4ms/step - loss:
0.0196 - mae: 0.1059 - val_loss: 0.0342 - val_mae: 0.1379
Epoch 170/1000
57/57            0s 4ms/step - loss:
0.0205 - mae: 0.1076 - val_loss: 0.0305 - val_mae: 0.1221
Epoch 171/1000
57/57            0s 4ms/step - loss:
0.0184 - mae: 0.1016 - val_loss: 0.0331 - val_mae: 0.1263
Epoch 172/1000
57/57            0s 4ms/step - loss:
0.0177 - mae: 0.1014 - val_loss: 0.0290 - val_mae: 0.1217
Epoch 173/1000
57/57            0s 4ms/step - loss:
0.0198 - mae: 0.1067 - val_loss: 0.0269 - val_mae: 0.1195
Epoch 174/1000
57/57            0s 4ms/step - loss:
0.0176 - mae: 0.1007 - val_loss: 0.0715 - val_mae: 0.2137
Epoch 175/1000
57/57            0s 4ms/step - loss:
0.0186 - mae: 0.1035 - val_loss: 0.0321 - val_mae: 0.1330
Epoch 176/1000
57/57            0s 4ms/step - loss:
0.0180 - mae: 0.1012 - val_loss: 0.1143 - val_mae: 0.2746
Epoch 177/1000
57/57            0s 4ms/step - loss:
0.0202 - mae: 0.1052 - val_loss: 0.0323 - val_mae: 0.1249
Epoch 178/1000
57/57            0s 4ms/step - loss:
0.0158 - mae: 0.0935 - val_loss: 0.0393 - val_mae: 0.1469
Epoch 179/1000
57/57            0s 4ms/step - loss:
0.0174 - mae: 0.1008 - val_loss: 0.0647 - val_mae: 0.1912
```

```
Epoch 180/1000
57/57             0s 4ms/step - loss:
0.0184 - mae: 0.0994 - val_loss: 0.0344 - val_mae: 0.1318
Epoch 181/1000
57/57             0s 4ms/step - loss:
0.0182 - mae: 0.1040 - val_loss: 0.0296 - val_mae: 0.1246
Epoch 182/1000
57/57             0s 4ms/step - loss:
0.0164 - mae: 0.0984 - val_loss: 0.0287 - val_mae: 0.1201
Epoch 183/1000
57/57             0s 4ms/step - loss:
0.0177 - mae: 0.0991 - val_loss: 0.0452 - val_mae: 0.1526
Epoch 184/1000
57/57             0s 4ms/step - loss:
0.0169 - mae: 0.1013 - val_loss: 0.0400 - val_mae: 0.1429
Epoch 185/1000
57/57             0s 4ms/step - loss:
0.0190 - mae: 0.1058 - val_loss: 0.0292 - val_mae: 0.1183
Epoch 186/1000
57/57             0s 4ms/step - loss:
0.0184 - mae: 0.1029 - val_loss: 0.0355 - val_mae: 0.1323
Epoch 186: early stopping
Restoring model weights from the end of the best epoch: 136.
Epoch 1/1000
46/46             0s 5ms/step - loss:
0.0220 - mae: 0.1076 - val_loss: 0.0128 - val_mae: 0.0874
Epoch 2/1000
46/46             0s 5ms/step - loss:
0.0263 - mae: 0.1248 - val_loss: 0.0505 - val_mae: 0.1731
Epoch 3/1000
46/46             0s 5ms/step - loss:
0.0225 - mae: 0.1138 - val_loss: 0.0119 - val_mae: 0.0865
Epoch 4/1000
46/46             0s 5ms/step - loss:
0.0209 - mae: 0.1093 - val_loss: 0.0234 - val_mae: 0.1132
Epoch 5/1000
46/46             0s 5ms/step - loss:
0.0226 - mae: 0.1155 - val_loss: 0.0171 - val_mae: 0.0978
Epoch 6/1000
46/46             0s 5ms/step - loss:
0.0251 - mae: 0.1172 - val_loss: 0.0123 - val_mae: 0.0840
Epoch 7/1000
46/46             0s 5ms/step - loss:
0.0189 - mae: 0.1001 - val_loss: 0.0294 - val_mae: 0.1309
Epoch 8/1000
46/46             0s 5ms/step - loss:
0.0222 - mae: 0.1095 - val_loss: 0.1270 - val_mae: 0.3016
Epoch 9/1000
```

```
46/46            0s 4ms/step - loss:
0.0246 - mae: 0.1188 - val_loss: 0.1011 - val_mae: 0.2752
Epoch 10/1000
46/46            0s 5ms/step - loss:
0.0195 - mae: 0.1067 - val_loss: 0.0423 - val_mae: 0.1665
Epoch 11/1000
46/46            0s 5ms/step - loss:
0.0212 - mae: 0.1088 - val_loss: 0.0358 - val_mae: 0.1518
Epoch 12/1000
46/46            0s 5ms/step - loss:
0.0236 - mae: 0.1155 - val_loss: 0.0136 - val_mae: 0.0854
Epoch 13/1000
46/46            0s 4ms/step - loss:
0.0218 - mae: 0.1065 - val_loss: 0.0136 - val_mae: 0.0893
Epoch 14/1000
46/46            0s 5ms/step - loss:
0.0188 - mae: 0.1038 - val_loss: 0.0148 - val_mae: 0.0914
Epoch 15/1000
46/46            0s 5ms/step - loss:
0.0203 - mae: 0.1084 - val_loss: 0.0436 - val_mae: 0.1657
Epoch 16/1000
46/46            0s 5ms/step - loss:
0.0235 - mae: 0.1136 - val_loss: 0.0305 - val_mae: 0.1366
Epoch 17/1000
46/46            0s 5ms/step - loss:
0.0206 - mae: 0.1067 - val_loss: 0.0746 - val_mae: 0.2195
Epoch 18/1000
46/46            0s 5ms/step - loss:
0.0197 - mae: 0.1058 - val_loss: 0.0239 - val_mae: 0.1244
Epoch 19/1000
46/46            0s 5ms/step - loss:
0.0193 - mae: 0.1039 - val_loss: 0.0750 - val_mae: 0.2353
Epoch 20/1000
46/46            0s 5ms/step - loss:
0.0209 - mae: 0.1102 - val_loss: 0.0141 - val_mae: 0.0901
Epoch 21/1000
46/46            0s 5ms/step - loss:
0.0217 - mae: 0.1079 - val_loss: 0.0173 - val_mae: 0.0971
Epoch 22/1000
46/46            0s 5ms/step - loss:
0.0175 - mae: 0.1014 - val_loss: 0.0282 - val_mae: 0.1329
Epoch 23/1000
46/46            0s 4ms/step - loss:
0.0206 - mae: 0.1099 - val_loss: 0.0173 - val_mae: 0.1014
Epoch 24/1000
46/46            0s 5ms/step - loss:
0.0212 - mae: 0.1098 - val_loss: 0.0237 - val_mae: 0.1235
Epoch 25/1000
```

```
46/46            0s 5ms/step - loss:
0.0185 - mae: 0.1040 - val_loss: 0.0454 - val_mae: 0.1716
Epoch 26/1000
46/46            0s 4ms/step - loss:
0.0181 - mae: 0.1039 - val_loss: 0.0163 - val_mae: 0.0949
Epoch 27/1000
46/46            0s 5ms/step - loss:
0.0202 - mae: 0.1018 - val_loss: 0.0162 - val_mae: 0.1002
Epoch 28/1000
46/46            0s 5ms/step - loss:
0.0176 - mae: 0.0979 - val_loss: 0.0146 - val_mae: 0.0897
Epoch 29/1000
46/46            0s 5ms/step - loss:
0.0219 - mae: 0.1083 - val_loss: 0.0147 - val_mae: 0.0976
Epoch 30/1000
46/46            0s 5ms/step - loss:
0.0192 - mae: 0.1019 - val_loss: 0.0186 - val_mae: 0.1061
Epoch 31/1000
46/46            0s 5ms/step - loss:
0.0176 - mae: 0.1005 - val_loss: 0.0281 - val_mae: 0.1325
Epoch 32/1000
46/46            0s 5ms/step - loss:
0.0201 - mae: 0.1079 - val_loss: 0.0347 - val_mae: 0.1378
Epoch 33/1000
46/46            0s 5ms/step - loss:
0.0172 - mae: 0.0965 - val_loss: 0.0168 - val_mae: 0.1001
Epoch 34/1000
46/46            0s 5ms/step - loss:
0.0210 - mae: 0.1069 - val_loss: 0.0151 - val_mae: 0.0910
Epoch 35/1000
46/46            0s 5ms/step - loss:
0.0176 - mae: 0.1020 - val_loss: 0.0138 - val_mae: 0.0899
Epoch 36/1000
46/46            0s 5ms/step - loss:
0.0205 - mae: 0.1099 - val_loss: 0.0213 - val_mae: 0.1133
Epoch 37/1000
46/46            0s 5ms/step - loss:
0.0174 - mae: 0.0990 - val_loss: 0.0569 - val_mae: 0.1936
Epoch 38/1000
46/46            0s 5ms/step - loss:
0.0167 - mae: 0.0990 - val_loss: 0.0223 - val_mae: 0.1156
Epoch 39/1000
46/46            0s 4ms/step - loss:
0.0182 - mae: 0.1022 - val_loss: 0.0186 - val_mae: 0.1072
Epoch 40/1000
46/46            0s 4ms/step - loss:
0.0208 - mae: 0.1108 - val_loss: 0.0192 - val_mae: 0.1082
Epoch 41/1000
```

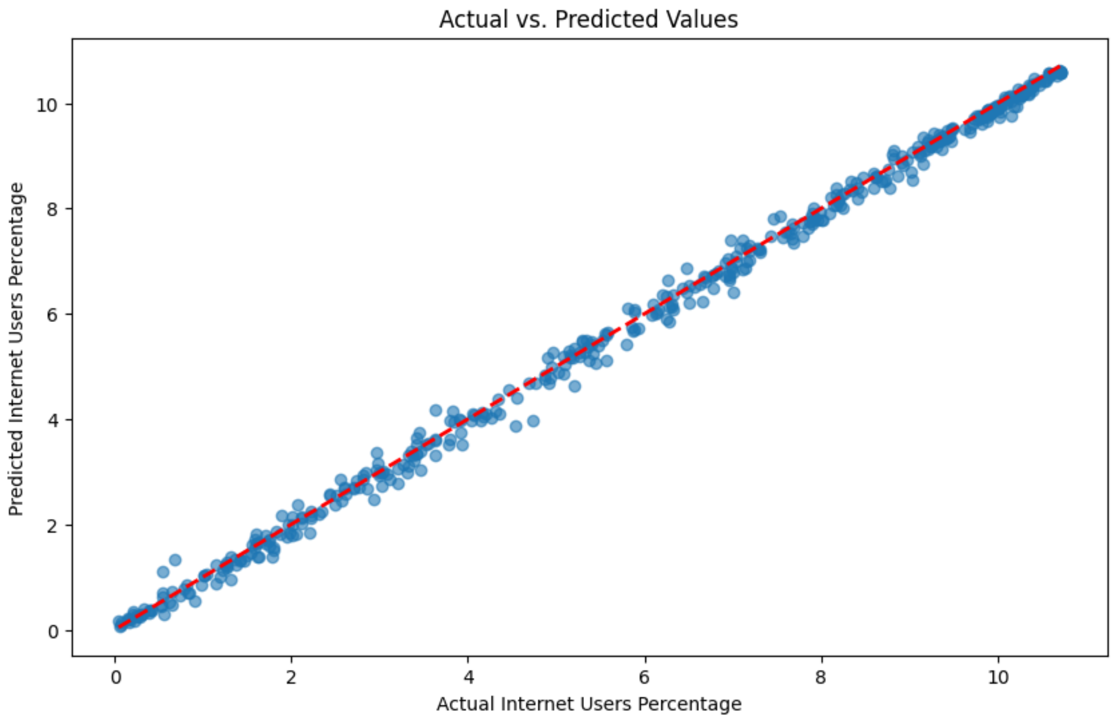
```
46/46            0s 5ms/step - loss:
0.0166 - mae: 0.0976 - val_loss: 0.0676 - val_mae: 0.2176
Epoch 42/1000
46/46            0s 5ms/step - loss:
0.0199 - mae: 0.1043 - val_loss: 0.0205 - val_mae: 0.1108
Epoch 43/1000
46/46            0s 5ms/step - loss:
0.0162 - mae: 0.0959 - val_loss: 0.0343 - val_mae: 0.1429
Epoch 44/1000
46/46            0s 5ms/step - loss:
0.0173 - mae: 0.0983 - val_loss: 0.0183 - val_mae: 0.1036
Epoch 45/1000
46/46            0s 5ms/step - loss:
0.0197 - mae: 0.1074 - val_loss: 0.0406 - val_mae: 0.1601
Epoch 46/1000
46/46            0s 5ms/step - loss:
0.0164 - mae: 0.0994 - val_loss: 0.0294 - val_mae: 0.1307
Epoch 47/1000
46/46            0s 5ms/step - loss:
0.0179 - mae: 0.0993 - val_loss: 0.0191 - val_mae: 0.1044
Epoch 48/1000
46/46            0s 5ms/step - loss:
0.0149 - mae: 0.0906 - val_loss: 0.0279 - val_mae: 0.1354
Epoch 49/1000
46/46            0s 5ms/step - loss:
0.0188 - mae: 0.1080 - val_loss: 0.0328 - val_mae: 0.1371
Epoch 50/1000
46/46            0s 5ms/step - loss:
0.0157 - mae: 0.0932 - val_loss: 0.0231 - val_mae: 0.1183
Epoch 51/1000
46/46            0s 5ms/step - loss:
0.0181 - mae: 0.1000 - val_loss: 0.0831 - val_mae: 0.2334
Epoch 52/1000
46/46            0s 4ms/step - loss:
0.0168 - mae: 0.0978 - val_loss: 0.0266 - val_mae: 0.1269
Epoch 53/1000
46/46            0s 5ms/step - loss:
0.0170 - mae: 0.0990 - val_loss: 0.0317 - val_mae: 0.1384
Epoch 53: early stopping
Restoring model weights from the end of the best epoch: 3.
Raw evaluate results: [0.031013615429401398, 0.13001972436904907]
```



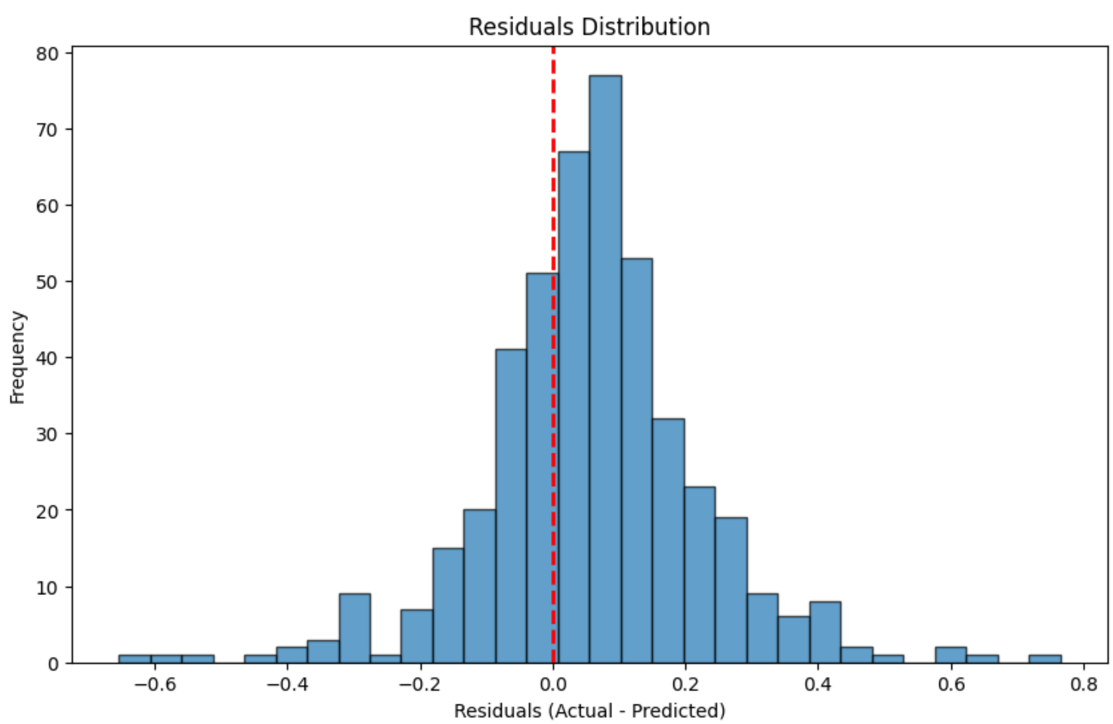
[6]:
```
# Predictions and Plots
predictions = model_tanh_rms.predict(test_features)

if not np.any(np.isnan(predictions)):
    predictions = predictions.flatten()
    final_tanh_results = pd.DataFrame({'Actual': test_target, 'Predicted':␣
↪predictions})

    # Scatter plot: Actual vs Predicted
    plt.figure(figsize=(10, 6))
    plt.scatter(final_tanh_results['Actual'], final_tanh_results['Predicted'],␣
↪alpha=0.6)
    plt.plot([min(final_tanh_results['Actual']),␣
↪max(final_tanh_results['Actual'])],
             [min(final_tanh_results['Actual']),␣
↪max(final_tanh_results['Actual'])],
             color='red', linestyle='--', lw=2)
    plt.title('Actual vs. Predicted Values')
    plt.xlabel('Actual Internet Users Percentage')
    plt.ylabel('Predicted Internet Users Percentage')
    plt.show()

    # Residual plot
```

```
    final_tanh_results['Residuals'] = final_tanh_results['Actual'] -␣
↪final_tanh_results['Predicted']
    plt.figure(figsize=(10, 6))
    plt.hist(final_tanh_results['Residuals'], bins=30, edgecolor='black',␣
↪alpha=0.7)
    plt.axvline(x=0, color='red', linestyle='--', lw=2)
    plt.title('Residuals Distribution')
    plt.xlabel('Residuals (Actual - Predicted)')
    plt.ylabel('Frequency')
    plt.show()

    # Training & validation MAE
    mae_history = history_tanh.history['mae']
    val_mae_history = history_tanh.history['val_mae']
    epochs_range = range(1, len(mae_history) + 1)

    plt.figure(figsize=(10, 6))
    plt.plot(epochs_range, mae_history, 'b', label='Training MAE')
    plt.plot(epochs_range, val_mae_history, 'r', label='Validation MAE')
    plt.title('Training and Validation MAE over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('MAE')
    plt.legend()
    plt.show()

else:
    print("Predictions contain invalid values (NaN or infinity). Cannot plot.")
```

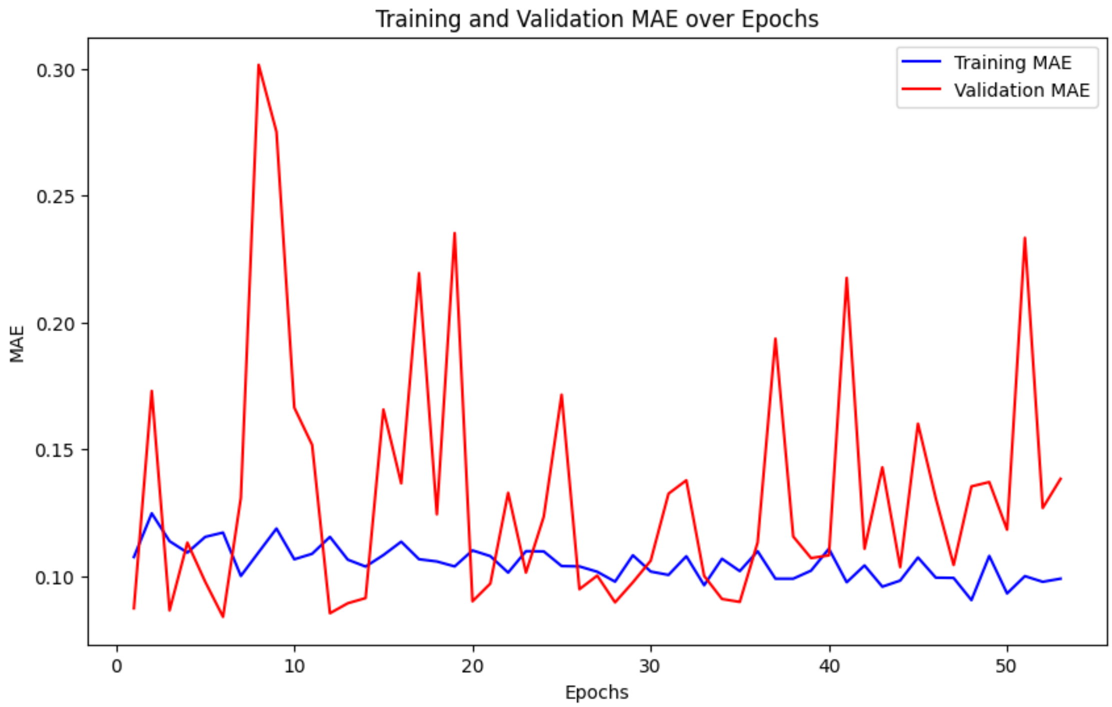
```
15/15              0s 5ms/step
```



Actual vs. Predicted Values
Predicted Internet Users Percentage
Actual Internet Users Percentage
0
2
4
6
8
10



Residuals Distribution
Frequency
Residuals (Actual - Predicted)
0
10
20
30
40
50
60
70
80
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8



[7]:

```
print("\n--- Training Model with Sigmoid and SGD ---")
model_sig = Sequential()
model_sig.add(Dense(256, input_dim=train_features.shape[1], activation='tanh'))
model_sig.add(Dense(128, activation='sigmoid'))
model_sig.add(Dense(64, activation='sigmoid'))
model_sig.add(Dense(1, activation='linear'))

model_sig.compile(optimizer=SGD(learning_rate=0.001), loss="mse",␣
↪metrics=["mae"])

# Define early stopping
early_stop_sig = EarlyStopping(monitor='val_loss', patience=50, verbose=1,␣
↪restore_best_weights=True)

model_sig.fit(train_features, train_target,
                    epochs=1000,
                    batch_size=32,
                    validation_data=(test_features, test_target),
                    callbacks=[early_stop_tanh],
                    verbose=1)

loss_sig = model_sig.evaluate(test_features, test_target, verbose=0)

history_sig = model_sig.fit(train_features, train_target,
```

```
                     epochs=1000,
                     batch_size=32,
                     validation_split=0.2,
                     callbacks=[early_stop_tanh],
                     verbose=1)

sig_results = model_sig.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", sig_results)

# Plot the training and validation MAE
mae_sig = history_sig.history['mae']
val_mae_sig = history_sig.history['val_mae']
epochs = range(1, len(mae_sig) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_sig, 'b', label='Training MAE')
plt.plot(epochs, val_mae_sig, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Sigmoid + SGD )')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()
```

```
--- Training Model with Sigmoid and SGD ---
Epoch 1/1000
```

```
C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
57/57                 1s 7ms/step - loss:
15.5124 - mae: 3.3635 - val_loss: 11.1005 - val_mae: 2.9514
Epoch 2/1000
57/57                 0s 4ms/step - loss:
10.5936 - mae: 2.8569 - val_loss: 10.9535 - val_mae: 2.9087
Epoch 3/1000
57/57                 0s 4ms/step - loss:
10.5046 - mae: 2.8326 - val_loss: 10.8898 - val_mae: 2.8966
Epoch 4/1000
57/57                 0s 4ms/step - loss:
10.4369 - mae: 2.8190 - val_loss: 10.7944 - val_mae: 2.8919
Epoch 5/1000
57/57                 0s 4ms/step - loss:
10.3666 - mae: 2.8107 - val_loss: 10.7107 - val_mae: 2.8837
Epoch 6/1000
57/57                 0s 4ms/step - loss:
10.2855 - mae: 2.8008 - val_loss: 10.6282 - val_mae: 2.8697
```

```
Epoch 7/1000
57/57            0s 4ms/step - loss:
10.2033 - mae: 2.7890 - val_loss: 10.5352 - val_mae: 2.8572
Epoch 8/1000
57/57            0s 4ms/step - loss:
10.1149 - mae: 2.7787 - val_loss: 10.4480 - val_mae: 2.8388
Epoch 9/1000
57/57            0s 4ms/step - loss:
10.0111 - mae: 2.7631 - val_loss: 10.3361 - val_mae: 2.8233
Epoch 10/1000
57/57            0s 4ms/step - loss:
9.8933 - mae: 2.7443 - val_loss: 10.2015 - val_mae: 2.8084
Epoch 11/1000
57/57            0s 4ms/step - loss:
9.7695 - mae: 2.7280 - val_loss: 10.0677 - val_mae: 2.7871
Epoch 12/1000
57/57            0s 4ms/step - loss:
9.6302 - mae: 2.7082 - val_loss: 9.9047 - val_mae: 2.7655
Epoch 13/1000
57/57            0s 4ms/step - loss:
9.4725 - mae: 2.6836 - val_loss: 9.7075 - val_mae: 2.7457
Epoch 14/1000
57/57            0s 4ms/step - loss:
9.2902 - mae: 2.6636 - val_loss: 9.5096 - val_mae: 2.7103
Epoch 15/1000
57/57            0s 4ms/step - loss:
9.0737 - mae: 2.6283 - val_loss: 9.2605 - val_mae: 2.6768
Epoch 16/1000
57/57            0s 4ms/step - loss:
8.8252 - mae: 2.5914 - val_loss: 8.9734 - val_mae: 2.6375
Epoch 17/1000
57/57            0s 4ms/step - loss:
8.5368 - mae: 2.5477 - val_loss: 8.6416 - val_mae: 2.5934
Epoch 18/1000
57/57            0s 4ms/step - loss:
8.2107 - mae: 2.5039 - val_loss: 8.2842 - val_mae: 2.5307
Epoch 19/1000
57/57            0s 4ms/step - loss:
7.8265 - mae: 2.4413 - val_loss: 7.8573 - val_mae: 2.4628
Epoch 20/1000
57/57            0s 4ms/step - loss:
7.3989 - mae: 2.3691 - val_loss: 7.3418 - val_mae: 2.3887
Epoch 21/1000
57/57            0s 4ms/step - loss:
6.8962 - mae: 2.2859 - val_loss: 6.7829 - val_mae: 2.3053
Epoch 22/1000
57/57            0s 4ms/step - loss:
6.3638 - mae: 2.2050 - val_loss: 6.1984 - val_mae: 2.1914
```

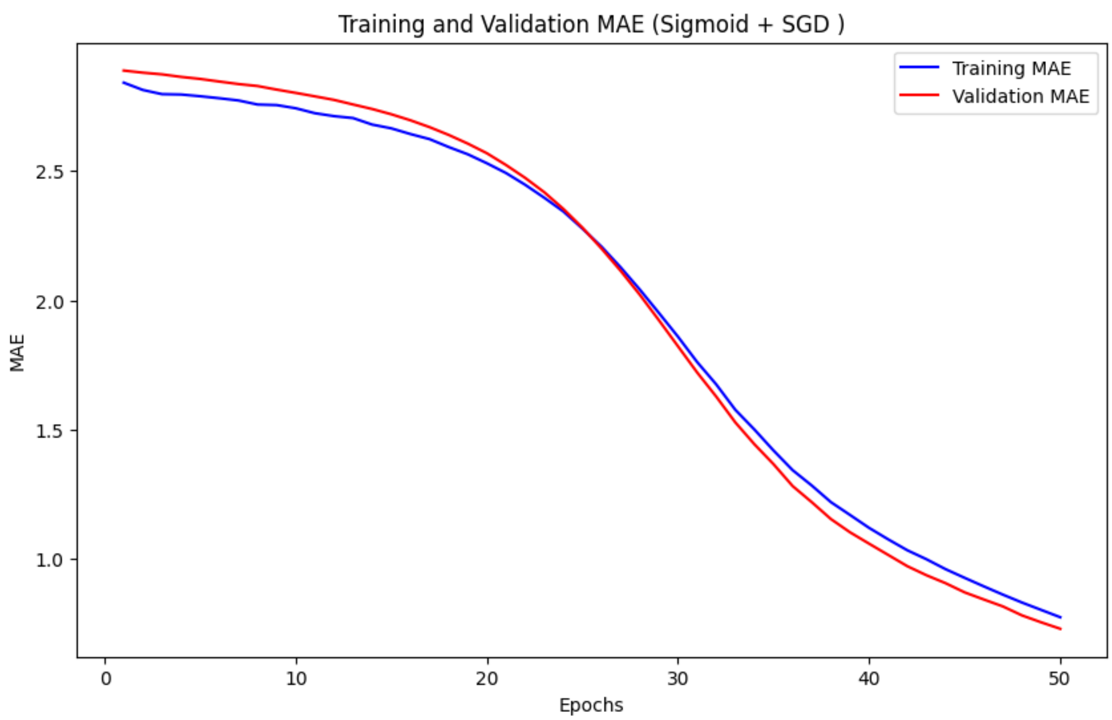
```
Epoch 23/1000
57/57          0s 4ms/step - loss:
5.7877 - mae: 2.0941 - val_loss: 5.5468 - val_mae: 2.0832
Epoch 24/1000
57/57          0s 4ms/step - loss:
5.1886 - mae: 1.9841 - val_loss: 4.9056 - val_mae: 1.9594
Epoch 25/1000
57/57          0s 4ms/step - loss:
4.5957 - mae: 1.8652 - val_loss: 4.2833 - val_mae: 1.8293
Epoch 26/1000
57/57          0s 4ms/step - loss:
4.0373 - mae: 1.7438 - val_loss: 3.7172 - val_mae: 1.6963
Epoch 27/1000
57/57          0s 4ms/step - loss:
3.5451 - mae: 1.6287 - val_loss: 3.2106 - val_mae: 1.5706
Epoch 28/1000
57/57          0s 4ms/step - loss:
3.1125 - mae: 1.5195 - val_loss: 2.7928 - val_mae: 1.4531
Epoch 29/1000
57/57          0s 4ms/step - loss:
2.7563 - mae: 1.4180 - val_loss: 2.4432 - val_mae: 1.3508
Epoch 30/1000
57/57          0s 4ms/step - loss:
2.4639 - mae: 1.3296 - val_loss: 2.1660 - val_mae: 1.2596
Epoch 31/1000
57/57          0s 4ms/step - loss:
2.2245 - mae: 1.2516 - val_loss: 1.9455 - val_mae: 1.1786
Epoch 32/1000
57/57          0s 4ms/step - loss:
2.0270 - mae: 1.1826 - val_loss: 1.7613 - val_mae: 1.1082
Epoch 33/1000
57/57          0s 4ms/step - loss:
1.8565 - mae: 1.1214 - val_loss: 1.6041 - val_mae: 1.0468
Epoch 34/1000
57/57          0s 4ms/step - loss:
1.7107 - mae: 1.0673 - val_loss: 1.4774 - val_mae: 0.9911
Epoch 35/1000
57/57          0s 4ms/step - loss:
1.5796 - mae: 1.0189 - val_loss: 1.3589 - val_mae: 0.9426
Epoch 36/1000
57/57          0s 4ms/step - loss:
1.4606 - mae: 0.9741 - val_loss: 1.2570 - val_mae: 0.8982
Epoch 37/1000
57/57          0s 4ms/step - loss:
1.3505 - mae: 0.9319 - val_loss: 1.1556 - val_mae: 0.8581
Epoch 38/1000
57/57          0s 4ms/step - loss:
1.2494 - mae: 0.8931 - val_loss: 1.0757 - val_mae: 0.8201
```

```
Epoch 39/1000
57/57            0s 4ms/step - loss:
1.1523 - mae: 0.8517 - val_loss: 0.9865 - val_mae: 0.7861
Epoch 40/1000
57/57            0s 4ms/step - loss:
1.0659 - mae: 0.8198 - val_loss: 0.9206 - val_mae: 0.7515
Epoch 41/1000
57/57            0s 4ms/step - loss:
0.9841 - mae: 0.7833 - val_loss: 0.8441 - val_mae: 0.7184
Epoch 42/1000
57/57            0s 4ms/step - loss:
0.9075 - mae: 0.7515 - val_loss: 0.7802 - val_mae: 0.6875
Epoch 43/1000
57/57            0s 4ms/step - loss:
0.8363 - mae: 0.7197 - val_loss: 0.7253 - val_mae: 0.6590
Epoch 44/1000
57/57            0s 4ms/step - loss:
0.7718 - mae: 0.6874 - val_loss: 0.6686 - val_mae: 0.6302
Epoch 45/1000
57/57            0s 4ms/step - loss:
0.7121 - mae: 0.6598 - val_loss: 0.6257 - val_mae: 0.6075
Epoch 46/1000
57/57            0s 4ms/step - loss:
0.6591 - mae: 0.6316 - val_loss: 0.5765 - val_mae: 0.5818
Epoch 47/1000
57/57            0s 4ms/step - loss:
0.6080 - mae: 0.6062 - val_loss: 0.5372 - val_mae: 0.5609
Epoch 48/1000
57/57            0s 4ms/step - loss:
0.5626 - mae: 0.5817 - val_loss: 0.5034 - val_mae: 0.5449
Epoch 49/1000
57/57            0s 4ms/step - loss:
0.5253 - mae: 0.5610 - val_loss: 0.4682 - val_mae: 0.5235
Epoch 50/1000
57/57            0s 4ms/step - loss:
0.4876 - mae: 0.5390 - val_loss: 0.4398 - val_mae: 0.5074
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Epoch 1/1000
46/46            0s 5ms/step - loss:
10.5247 - mae: 2.8421 - val_loss: 11.0286 - val_mae: 2.8889
Epoch 2/1000
46/46            0s 4ms/step - loss:
10.4448 - mae: 2.8138 - val_loss: 10.9504 - val_mae: 2.8807
Epoch 3/1000
46/46            0s 4ms/step - loss:
10.3760 - mae: 2.7976 - val_loss: 10.8003 - val_mae: 2.8742
Epoch 4/1000
```

```
46/46            0s 4ms/step - loss:
10.3283 - mae: 2.7962 - val_loss: 10.7750 - val_mae: 2.8644
Epoch 5/1000
46/46            0s 4ms/step - loss:
10.2696 - mae: 2.7896 - val_loss: 10.6856 - val_mae: 2.8565
Epoch 6/1000
46/46            0s 4ms/step - loss:
10.2060 - mae: 2.7818 - val_loss: 10.6512 - val_mae: 2.8464
Epoch 7/1000
46/46            0s 4ms/step - loss:
10.1368 - mae: 2.7733 - val_loss: 10.6259 - val_mae: 2.8367
Epoch 8/1000
46/46            0s 4ms/step - loss:
10.0725 - mae: 2.7576 - val_loss: 10.4373 - val_mae: 2.8291
Epoch 9/1000
46/46            0s 4ms/step - loss:
9.9949 - mae: 2.7554 - val_loss: 10.4078 - val_mae: 2.8152
Epoch 10/1000
46/46            0s 4ms/step - loss:
9.9114 - mae: 2.7431 - val_loss: 10.3590 - val_mae: 2.8025
Epoch 11/1000
46/46            0s 4ms/step - loss:
9.8299 - mae: 2.7240 - val_loss: 10.2413 - val_mae: 2.7891
Epoch 12/1000
46/46            0s 4ms/step - loss:
9.7392 - mae: 2.7126 - val_loss: 10.0781 - val_mae: 2.7755
Epoch 13/1000
46/46            0s 4ms/step - loss:
9.6296 - mae: 2.7051 - val_loss: 10.0677 - val_mae: 2.7576
Epoch 14/1000
46/46            0s 4ms/step - loss:
9.5125 - mae: 2.6799 - val_loss: 9.8543 - val_mae: 2.7402
Epoch 15/1000
46/46            0s 4ms/step - loss:
9.3849 - mae: 2.6654 - val_loss: 9.6972 - val_mae: 2.7198
Epoch 16/1000
46/46            0s 4ms/step - loss:
9.2420 - mae: 2.6434 - val_loss: 9.5490 - val_mae: 2.6965
Epoch 17/1000
46/46            0s 4ms/step - loss:
9.0848 - mae: 2.6238 - val_loss: 9.3772 - val_mae: 2.6701
Epoch 18/1000
46/46            0s 4ms/step - loss:
8.8992 - mae: 2.5931 - val_loss: 9.1860 - val_mae: 2.6400
Epoch 19/1000
46/46            0s 4ms/step - loss:
8.7000 - mae: 2.5651 - val_loss: 8.9386 - val_mae: 2.6064
Epoch 20/1000
```

```
46/46            0s 4ms/step - loss:
8.4768 - mae: 2.5304 - val_loss: 8.6319 - val_mae: 2.5691
Epoch 21/1000
46/46            0s 4ms/step - loss:
8.2203 - mae: 2.4926 - val_loss: 8.3781 - val_mae: 2.5236
Epoch 22/1000
46/46            0s 4ms/step - loss:
7.9340 - mae: 2.4472 - val_loss: 8.0351 - val_mae: 2.4739
Epoch 23/1000
46/46            0s 4ms/step - loss:
7.6160 - mae: 2.3971 - val_loss: 7.6442 - val_mae: 2.4184
Epoch 24/1000
46/46            0s 4ms/step - loss:
7.2628 - mae: 2.3440 - val_loss: 7.2563 - val_mae: 2.3539
Epoch 25/1000
46/46            0s 4ms/step - loss:
6.8748 - mae: 2.2775 - val_loss: 6.8506 - val_mae: 2.2811
Epoch 26/1000
46/46            0s 4ms/step - loss:
6.4587 - mae: 2.2076 - val_loss: 6.4022 - val_mae: 2.2003
Epoch 27/1000
46/46            0s 4ms/step - loss:
6.0142 - mae: 2.1278 - val_loss: 5.9527 - val_mae: 2.1143
Epoch 28/1000
46/46            0s 4ms/step - loss:
5.5499 - mae: 2.0421 - val_loss: 5.4437 - val_mae: 2.0220
Epoch 29/1000
46/46            0s 4ms/step - loss:
5.0816 - mae: 1.9515 - val_loss: 4.8996 - val_mae: 1.9233
Epoch 30/1000
46/46            0s 4ms/step - loss:
4.6139 - mae: 1.8595 - val_loss: 4.4079 - val_mae: 1.8224
Epoch 31/1000
46/46            0s 4ms/step - loss:
4.1711 - mae: 1.7617 - val_loss: 3.8937 - val_mae: 1.7214
Epoch 32/1000
46/46            0s 4ms/step - loss:
3.7566 - mae: 1.6751 - val_loss: 3.6018 - val_mae: 1.6279
Epoch 33/1000
46/46            0s 4ms/step - loss:
3.3939 - mae: 1.5767 - val_loss: 3.1314 - val_mae: 1.5280
Epoch 34/1000
46/46            0s 4ms/step - loss:
3.0645 - mae: 1.5004 - val_loss: 2.8644 - val_mae: 1.4434
Epoch 35/1000
46/46            0s 4ms/step - loss:
2.7844 - mae: 1.4193 - val_loss: 2.6353 - val_mae: 1.3664
Epoch 36/1000
```

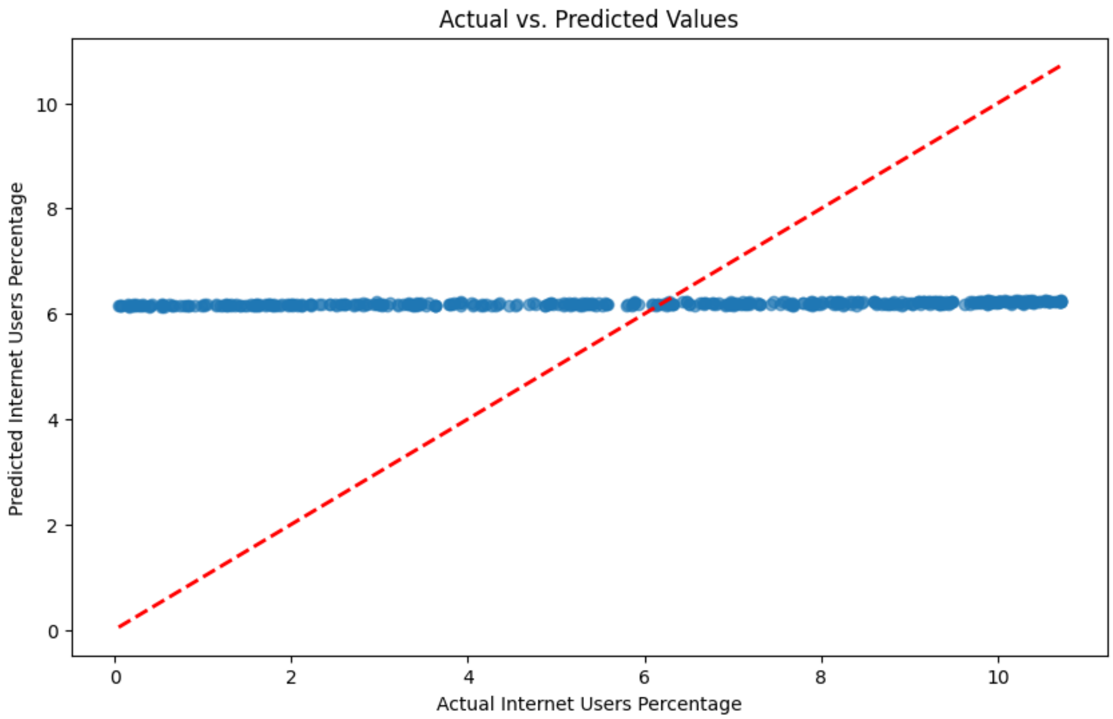
```
46/46             0s 4ms/step - loss:
2.5455 - mae: 1.3433 - val_loss: 2.2878 - val_mae: 1.2817
Epoch 37/1000
46/46             0s 4ms/step - loss:
2.3362 - mae: 1.2839 - val_loss: 2.1428 - val_mae: 1.2195
Epoch 38/1000
46/46             0s 4ms/step - loss:
2.1570 - mae: 1.2197 - val_loss: 1.9062 - val_mae: 1.1543
Epoch 39/1000
46/46             0s 4ms/step - loss:
2.0037 - mae: 1.1702 - val_loss: 1.7780 - val_mae: 1.1031
Epoch 40/1000
46/46             0s 4ms/step - loss:
1.8652 - mae: 1.1201 - val_loss: 1.6814 - val_mae: 1.0590
Epoch 41/1000
46/46             0s 4ms/step - loss:
1.7424 - mae: 1.0757 - val_loss: 1.5526 - val_mae: 1.0155
Epoch 42/1000
46/46             0s 4ms/step - loss:
1.6296 - mae: 1.0335 - val_loss: 1.4062 - val_mae: 0.9720
Epoch 43/1000
46/46             0s 4ms/step - loss:
1.5291 - mae: 0.9988 - val_loss: 1.3329 - val_mae: 0.9370
Epoch 44/1000
46/46             0s 4ms/step - loss:
1.4305 - mae: 0.9605 - val_loss: 1.2692 - val_mae: 0.9062
Epoch 45/1000
46/46             0s 4ms/step - loss:
1.3424 - mae: 0.9269 - val_loss: 1.1731 - val_mae: 0.8707
Epoch 46/1000
46/46             0s 4ms/step - loss:
1.2557 - mae: 0.8940 - val_loss: 1.1186 - val_mae: 0.8432
Epoch 47/1000
46/46             0s 4ms/step - loss:
1.1765 - mae: 0.8620 - val_loss: 1.0621 - val_mae: 0.8163
Epoch 48/1000
46/46             0s 4ms/step - loss:
1.1009 - mae: 0.8314 - val_loss: 0.9632 - val_mae: 0.7816
Epoch 49/1000
46/46             0s 4ms/step - loss:
1.0309 - mae: 0.8030 - val_loss: 0.9058 - val_mae: 0.7550
Epoch 50/1000
46/46             0s 4ms/step - loss:
0.9641 - mae: 0.7746 - val_loss: 0.8555 - val_mae: 0.7299
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Raw evaluate results: [11.006515502929688, 2.9048659801483154]
```

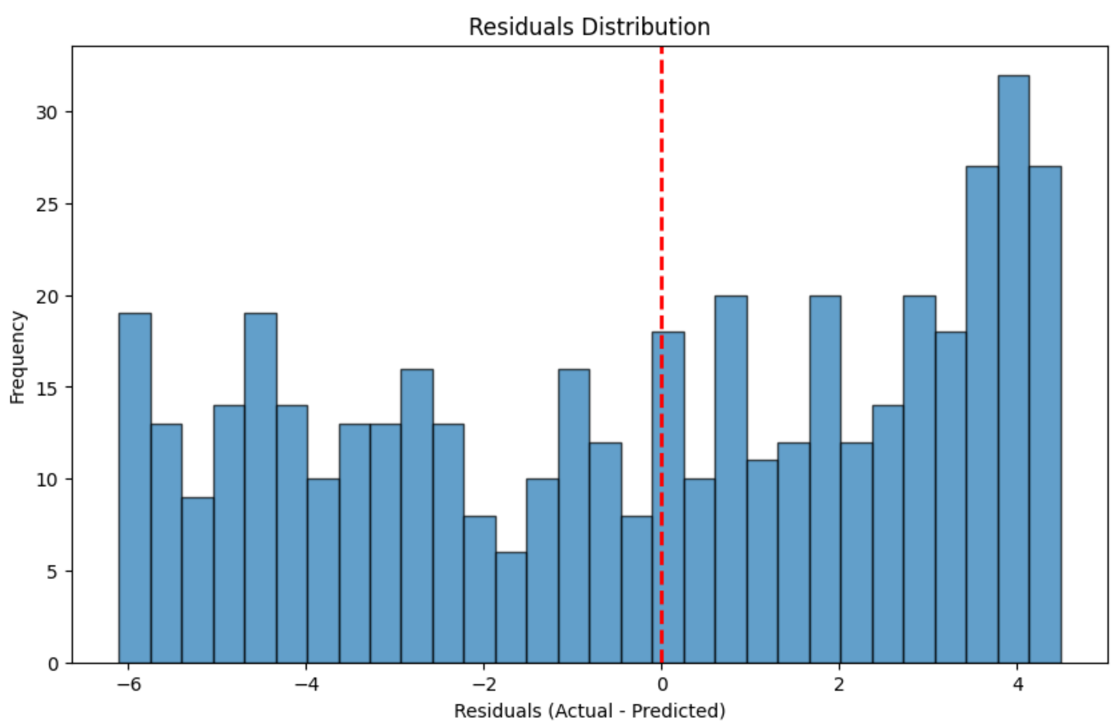


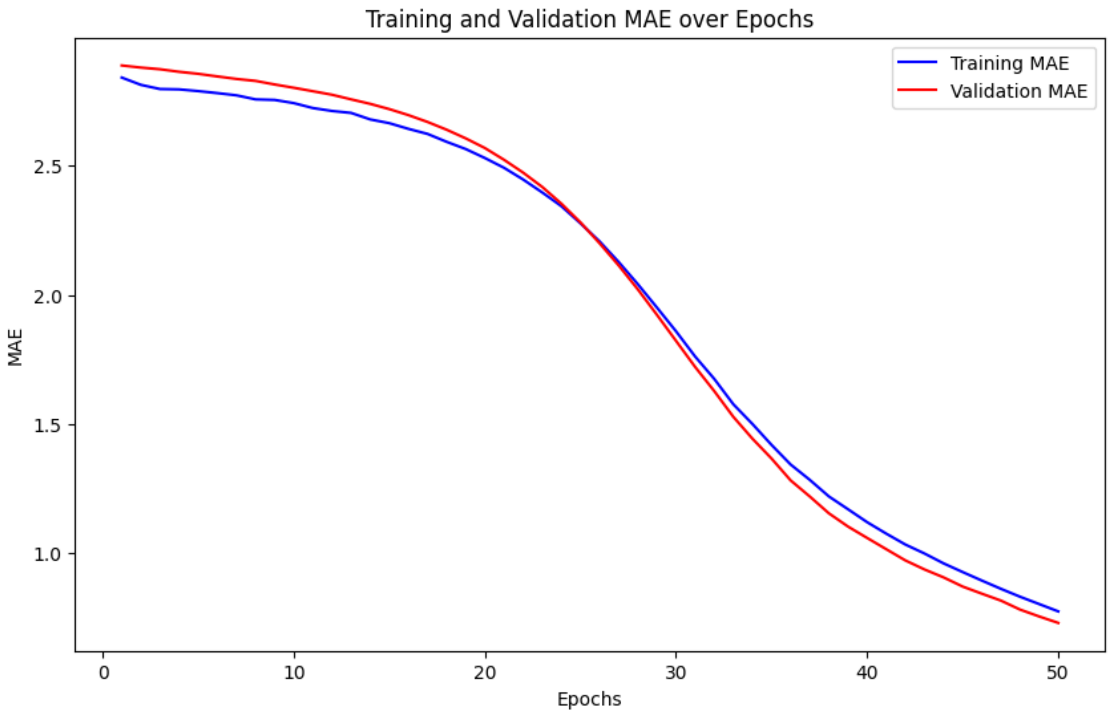
[9]:
```
# Predictions and Plots
sig_predictions = model_sig.predict(test_features)

if not np.any(np.isnan(sig_predictions)):
    predictions = sig_predictions.flatten()
    final_sig_results = pd.DataFrame({'Actual': test_target, 'Predicted':␣
↪predictions})

    # Scatter plot: Actual vs Predicted
    plt.figure(figsize=(10, 6))
    plt.scatter(final_sig_results['Actual'],final_sig_results['Predicted'],␣
↪alpha=0.6)
    plt.plot([min(final_sig_results['Actual']),␣
↪max(final_sig_results['Actual'])],
             [min(final_sig_results['Actual']),␣
↪max(final_sig_results['Actual'])],
             color='red', linestyle='--', lw=2)
    plt.title('Actual vs. Predicted Values')
    plt.xlabel('Actual Internet Users Percentage')
    plt.ylabel('Predicted Internet Users Percentage')
    plt.show()

    # Residual plot
```

```
    final_sig_results['Residuals'] = final_sig_results['Actual'] -␣
↪final_sig_results['Predicted']
    plt.figure(figsize=(10, 6))
    plt.hist(final_sig_results['Residuals'], bins=30, edgecolor='black',␣
↪alpha=0.7)
    plt.axvline(x=0, color='red', linestyle='--', lw=2)
    plt.title('Residuals Distribution')
    plt.xlabel('Residuals (Actual - Predicted)')
    plt.ylabel('Frequency')
    plt.show()

    # Training & validation MAE
    mae_sig = history_sig.history['mae']
    val_mae__sig_history = history_sig.history['val_mae']
    epochs_range = range(1, len(mae_sig) + 1)

    plt.figure(figsize=(10, 6))
    plt.plot(epochs_range, mae_sig, 'b', label='Training MAE')
    plt.plot(epochs_range, val_mae__sig_history, 'r', label='Validation MAE')
    plt.title('Training and Validation MAE over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('MAE')
    plt.legend()
    plt.show()

else:
    print("Predictions contain invalid values (NaN or infinity). Cannot plot.")
```

```
15/15              0s 3ms/step
```


Actual vs. Predicted Values
Predicted Internet Users Percentage
Actual Internet Users Percentage
0
2
4
6
8
10
0
2
4
6
8
10


Residuals Distribution
Frequency
Residuals (Actual - Predicted)
0
5
10
15
20
25
30
−6
−4
−2
0
2
4



[10]:
```
print("\n--- Training Model with ReLu and Adam ---")
model_relu = Sequential()
model_relu.add(Dense(256, input_dim=train_features.shape[1], activation='relu'))
model_relu.add(Dense(128, activation='relu'))
model_relu.add(Dense(64, activation='relu'))
model_relu.add(Dense(1, activation='linear'))

model_relu.compile(optimizer=Adam(learning_rate=0.001), loss="mse",␣
 ↪metrics=["mae"])

# Define early stopping
early_stop_relu = EarlyStopping(monitor='val_loss', patience=50, verbose=1,␣
 ↪restore_best_weights=True)

model_relu.fit(train_features, train_target,
                    epochs=1000,
                    batch_size=32,
                    validation_data=(test_features, test_target),
                    callbacks=[early_stop_tanh],
                    verbose=1)

loss_relu = model_relu.evaluate(test_features, test_target, verbose=0)
```

```
history_relu = model_relu.fit(train_features, train_target,
                    epochs=1000,
                    batch_size=32,
                    validation_split=0.2,
                    callbacks=[early_stop_tanh],
                    verbose=1)

relu_results = model_relu.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", relu_results)

# Plot the training and validation MAE
mae_relu = history_relu.history['mae']
val_mae_relu = history_relu.history['val_mae']
epochs = range(1, len(mae_relu) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_relu, 'b', label='Training MAE')
plt.plot(epochs, val_mae_relu, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Sigmoid + SGD )')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()
```

```
--- Training Model with ReLu and Adam ---
Epoch 1/1000
```

```
C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
57/57                  2s 7ms/step - loss:
8.6508 - mae: 2.1194 - val_loss: 1.1870 - val_mae: 0.8512
Epoch 2/1000
57/57                  0s 4ms/step - loss:
0.4331 - mae: 0.4755 - val_loss: 0.1411 - val_mae: 0.2912
Epoch 3/1000
57/57                  0s 4ms/step - loss:
0.1197 - mae: 0.2657 - val_loss: 0.0965 - val_mae: 0.2370
Epoch 4/1000
57/57                  0s 4ms/step - loss:
0.0837 - mae: 0.2202 - val_loss: 0.0759 - val_mae: 0.2038
Epoch 5/1000
57/57                  0s 4ms/step - loss:
0.0712 - mae: 0.2041 - val_loss: 0.0770 - val_mae: 0.2102
Epoch 6/1000
57/57                  0s 4ms/step - loss:
```

```
0.0610 - mae: 0.1884 - val_loss: 0.0736 - val_mae: 0.2003
Epoch 7/1000
57/57            0s 4ms/step - loss:
0.0588 - mae: 0.1820 - val_loss: 0.1048 - val_mae: 0.2538
Epoch 8/1000
57/57            0s 4ms/step - loss:
0.0585 - mae: 0.1843 - val_loss: 0.0644 - val_mae: 0.1862
Epoch 9/1000
57/57            0s 4ms/step - loss:
0.0606 - mae: 0.1878 - val_loss: 0.0675 - val_mae: 0.1940
Epoch 10/1000
57/57            0s 4ms/step - loss:
0.0496 - mae: 0.1649 - val_loss: 0.0611 - val_mae: 0.1819
Epoch 11/1000
57/57            0s 4ms/step - loss:
0.0489 - mae: 0.1666 - val_loss: 0.0570 - val_mae: 0.1802
Epoch 12/1000
57/57            0s 4ms/step - loss:
0.0429 - mae: 0.1518 - val_loss: 0.0672 - val_mae: 0.1975
Epoch 13/1000
57/57            0s 4ms/step - loss:
0.0456 - mae: 0.1620 - val_loss: 0.0578 - val_mae: 0.1778
Epoch 14/1000
57/57            0s 4ms/step - loss:
0.0432 - mae: 0.1569 - val_loss: 0.0639 - val_mae: 0.1914
Epoch 15/1000
57/57            0s 4ms/step - loss:
0.0396 - mae: 0.1466 - val_loss: 0.0574 - val_mae: 0.1733
Epoch 16/1000
57/57            0s 4ms/step - loss:
0.0419 - mae: 0.1513 - val_loss: 0.0541 - val_mae: 0.1725
Epoch 17/1000
57/57            0s 4ms/step - loss:
0.0362 - mae: 0.1411 - val_loss: 0.0551 - val_mae: 0.1695
Epoch 18/1000
57/57            0s 4ms/step - loss:
0.0382 - mae: 0.1468 - val_loss: 0.0502 - val_mae: 0.1642
Epoch 19/1000
57/57            0s 4ms/step - loss:
0.0420 - mae: 0.1549 - val_loss: 0.0504 - val_mae: 0.1618
Epoch 20/1000
57/57            0s 4ms/step - loss:
0.0337 - mae: 0.1366 - val_loss: 0.0526 - val_mae: 0.1617
Epoch 21/1000
57/57            0s 4ms/step - loss:
0.0325 - mae: 0.1324 - val_loss: 0.0451 - val_mae: 0.1527
Epoch 22/1000
57/57            0s 4ms/step - loss:
```

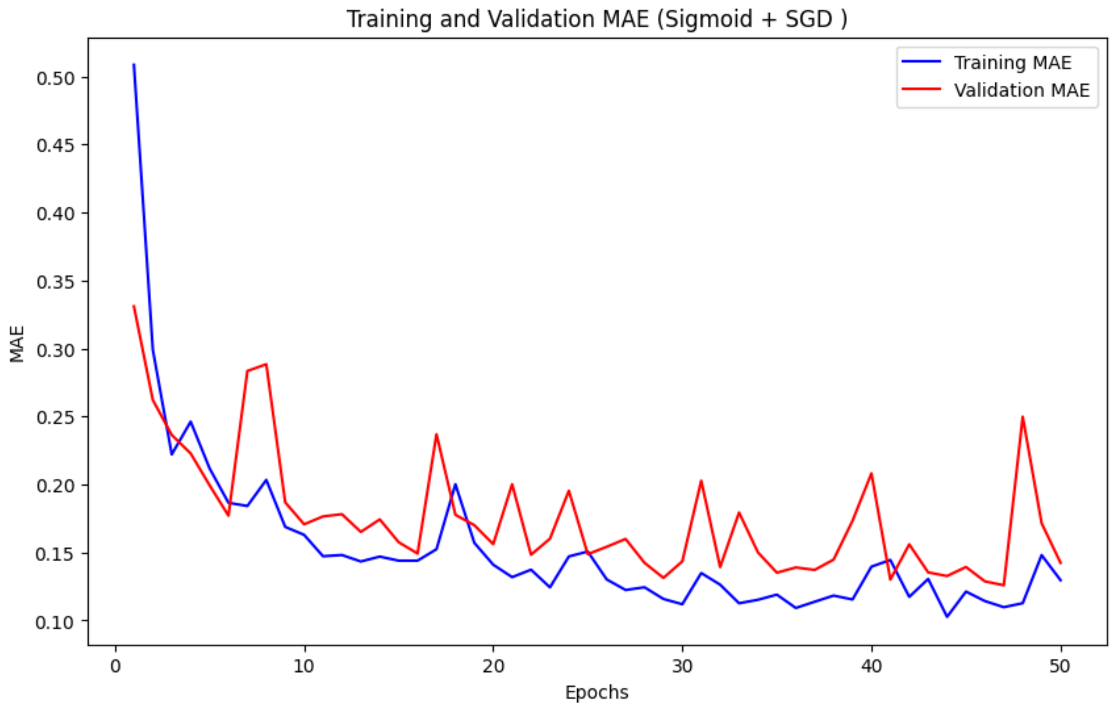
```
0.0384 - mae: 0.1479 - val_loss: 0.0556 - val_mae: 0.1764
Epoch 23/1000
57/57            0s 4ms/step - loss:
0.0306 - mae: 0.1286 - val_loss: 0.0563 - val_mae: 0.1762
Epoch 24/1000
57/57            0s 4ms/step - loss:
0.0394 - mae: 0.1501 - val_loss: 0.0546 - val_mae: 0.1698
Epoch 25/1000
57/57            0s 4ms/step - loss:
0.0346 - mae: 0.1389 - val_loss: 0.0478 - val_mae: 0.1656
Epoch 26/1000
57/57            0s 4ms/step - loss:
0.0338 - mae: 0.1387 - val_loss: 0.0449 - val_mae: 0.1486
Epoch 27/1000
57/57            0s 4ms/step - loss:
0.0328 - mae: 0.1321 - val_loss: 0.0457 - val_mae: 0.1565
Epoch 28/1000
57/57            0s 4ms/step - loss:
0.0320 - mae: 0.1345 - val_loss: 0.0439 - val_mae: 0.1499
Epoch 29/1000
57/57            0s 4ms/step - loss:
0.0317 - mae: 0.1323 - val_loss: 0.0558 - val_mae: 0.1830
Epoch 30/1000
57/57            0s 4ms/step - loss:
0.0272 - mae: 0.1237 - val_loss: 0.0414 - val_mae: 0.1503
Epoch 31/1000
57/57            0s 4ms/step - loss:
0.0266 - mae: 0.1214 - val_loss: 0.0453 - val_mae: 0.1539
Epoch 32/1000
57/57            0s 4ms/step - loss:
0.0277 - mae: 0.1251 - val_loss: 0.0454 - val_mae: 0.1548
Epoch 33/1000
57/57            0s 4ms/step - loss:
0.0266 - mae: 0.1180 - val_loss: 0.0466 - val_mae: 0.1614
Epoch 34/1000
57/57            0s 4ms/step - loss:
0.0252 - mae: 0.1165 - val_loss: 0.0385 - val_mae: 0.1417
Epoch 35/1000
57/57            0s 4ms/step - loss:
0.0326 - mae: 0.1379 - val_loss: 0.0460 - val_mae: 0.1586
Epoch 36/1000
57/57            0s 4ms/step - loss:
0.0242 - mae: 0.1165 - val_loss: 0.0364 - val_mae: 0.1397
Epoch 37/1000
57/57            0s 4ms/step - loss:
0.0259 - mae: 0.1178 - val_loss: 0.0345 - val_mae: 0.1326
Epoch 38/1000
57/57            0s 4ms/step - loss:
```

```
0.0216 - mae: 0.1066 - val_loss: 0.0359 - val_mae: 0.1335
Epoch 39/1000
57/57            0s 4ms/step - loss:
0.0245 - mae: 0.1134 - val_loss: 0.0385 - val_mae: 0.1407
Epoch 40/1000
57/57            0s 4ms/step - loss:
0.0283 - mae: 0.1272 - val_loss: 0.0355 - val_mae: 0.1368
Epoch 41/1000
57/57            0s 4ms/step - loss:
0.0221 - mae: 0.1116 - val_loss: 0.0387 - val_mae: 0.1450
Epoch 42/1000
57/57            0s 4ms/step - loss:
0.0257 - mae: 0.1212 - val_loss: 0.0384 - val_mae: 0.1467
Epoch 43/1000
57/57            0s 4ms/step - loss:
0.0236 - mae: 0.1118 - val_loss: 0.0310 - val_mae: 0.1259
Epoch 44/1000
57/57            0s 4ms/step - loss:
0.0256 - mae: 0.1191 - val_loss: 0.0360 - val_mae: 0.1404
Epoch 45/1000
57/57            0s 4ms/step - loss:
0.0260 - mae: 0.1224 - val_loss: 0.0551 - val_mae: 0.1914
Epoch 46/1000
57/57            0s 4ms/step - loss:
0.0378 - mae: 0.1529 - val_loss: 0.0406 - val_mae: 0.1501
Epoch 47/1000
57/57            0s 4ms/step - loss:
0.0242 - mae: 0.1164 - val_loss: 0.0378 - val_mae: 0.1437
Epoch 48/1000
57/57            0s 4ms/step - loss:
0.0202 - mae: 0.1068 - val_loss: 0.0283 - val_mae: 0.1186
Epoch 49/1000
57/57            0s 4ms/step - loss:
0.0230 - mae: 0.1144 - val_loss: 0.0358 - val_mae: 0.1405
Epoch 50/1000
57/57            0s 4ms/step - loss:
0.0227 - mae: 0.1131 - val_loss: 0.0367 - val_mae: 0.1419
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Epoch 1/1000
46/46            0s 5ms/step - loss:
0.4728 - mae: 0.5086 - val_loss: 0.1845 - val_mae: 0.3310
Epoch 2/1000
46/46            0s 5ms/step - loss:
0.1488 - mae: 0.2990 - val_loss: 0.1082 - val_mae: 0.2621
Epoch 3/1000
46/46            0s 5ms/step - loss:
0.0857 - mae: 0.2221 - val_loss: 0.0893 - val_mae: 0.2364
```

```
Epoch 4/1000
46/46            0s 5ms/step - loss:
0.1007 - mae: 0.2462 - val_loss: 0.0859 - val_mae: 0.2228
Epoch 5/1000
46/46            0s 5ms/step - loss:
0.0770 - mae: 0.2119 - val_loss: 0.0701 - val_mae: 0.1994
Epoch 6/1000
46/46            0s 5ms/step - loss:
0.0619 - mae: 0.1865 - val_loss: 0.0572 - val_mae: 0.1771
Epoch 7/1000
46/46            0s 5ms/step - loss:
0.0594 - mae: 0.1842 - val_loss: 0.1211 - val_mae: 0.2834
Epoch 8/1000
46/46            0s 5ms/step - loss:
0.0697 - mae: 0.2034 - val_loss: 0.1205 - val_mae: 0.2884
Epoch 9/1000
46/46            0s 4ms/step - loss:
0.0504 - mae: 0.1688 - val_loss: 0.0620 - val_mae: 0.1869
Epoch 10/1000
46/46            0s 4ms/step - loss:
0.0469 - mae: 0.1630 - val_loss: 0.0533 - val_mae: 0.1707
Epoch 11/1000
46/46            0s 4ms/step - loss:
0.0410 - mae: 0.1472 - val_loss: 0.0547 - val_mae: 0.1765
Epoch 12/1000
46/46            0s 4ms/step - loss:
0.0410 - mae: 0.1481 - val_loss: 0.0575 - val_mae: 0.1782
Epoch 13/1000
46/46            0s 4ms/step - loss:
0.0375 - mae: 0.1434 - val_loss: 0.0513 - val_mae: 0.1651
Epoch 14/1000
46/46            0s 5ms/step - loss:
0.0398 - mae: 0.1470 - val_loss: 0.0571 - val_mae: 0.1743
Epoch 15/1000
46/46            0s 4ms/step - loss:
0.0380 - mae: 0.1440 - val_loss: 0.0461 - val_mae: 0.1576
Epoch 16/1000
46/46            0s 4ms/step - loss:
0.0377 - mae: 0.1440 - val_loss: 0.0449 - val_mae: 0.1492
Epoch 17/1000
46/46            0s 5ms/step - loss:
0.0411 - mae: 0.1524 - val_loss: 0.0833 - val_mae: 0.2369
Epoch 18/1000
46/46            0s 5ms/step - loss:
0.0671 - mae: 0.2002 - val_loss: 0.0556 - val_mae: 0.1777
Epoch 19/1000
46/46            0s 5ms/step - loss:
0.0426 - mae: 0.1571 - val_loss: 0.0547 - val_mae: 0.1701
```

```
Epoch 20/1000
46/46            0s 4ms/step - loss:
0.0349 - mae: 0.1410 - val_loss: 0.0468 - val_mae: 0.1561
Epoch 21/1000
46/46            0s 5ms/step - loss:
0.0321 - mae: 0.1319 - val_loss: 0.0687 - val_mae: 0.2002
Epoch 22/1000
46/46            0s 5ms/step - loss:
0.0331 - mae: 0.1373 - val_loss: 0.0413 - val_mae: 0.1484
Epoch 23/1000
46/46            0s 4ms/step - loss:
0.0291 - mae: 0.1244 - val_loss: 0.0476 - val_mae: 0.1601
Epoch 24/1000
46/46            0s 4ms/step - loss:
0.0376 - mae: 0.1471 - val_loss: 0.0660 - val_mae: 0.1954
Epoch 25/1000
46/46            0s 4ms/step - loss:
0.0395 - mae: 0.1507 - val_loss: 0.0422 - val_mae: 0.1488
Epoch 26/1000
46/46            0s 5ms/step - loss:
0.0305 - mae: 0.1302 - val_loss: 0.0434 - val_mae: 0.1542
Epoch 27/1000
46/46            0s 5ms/step - loss:
0.0277 - mae: 0.1224 - val_loss: 0.0445 - val_mae: 0.1600
Epoch 28/1000
46/46            0s 5ms/step - loss:
0.0287 - mae: 0.1244 - val_loss: 0.0399 - val_mae: 0.1424
Epoch 29/1000
46/46            0s 4ms/step - loss:
0.0250 - mae: 0.1158 - val_loss: 0.0346 - val_mae: 0.1313
Epoch 30/1000
46/46            0s 5ms/step - loss:
0.0236 - mae: 0.1120 - val_loss: 0.0413 - val_mae: 0.1435
Epoch 31/1000
46/46            0s 5ms/step - loss:
0.0306 - mae: 0.1349 - val_loss: 0.0686 - val_mae: 0.2028
Epoch 32/1000
46/46            0s 4ms/step - loss:
0.0293 - mae: 0.1263 - val_loss: 0.0408 - val_mae: 0.1393
Epoch 33/1000
46/46            0s 5ms/step - loss:
0.0250 - mae: 0.1127 - val_loss: 0.0535 - val_mae: 0.1794
Epoch 34/1000
46/46            0s 4ms/step - loss:
0.0246 - mae: 0.1152 - val_loss: 0.0419 - val_mae: 0.1500
Epoch 35/1000
46/46            0s 4ms/step - loss:
0.0258 - mae: 0.1190 - val_loss: 0.0378 - val_mae: 0.1352
```

```
Epoch 36/1000
46/46            0s 5ms/step - loss:
0.0223 - mae: 0.1092 - val_loss: 0.0392 - val_mae: 0.1390
Epoch 37/1000
46/46            0s 4ms/step - loss:
0.0237 - mae: 0.1138 - val_loss: 0.0388 - val_mae: 0.1372
Epoch 38/1000
46/46            0s 4ms/step - loss:
0.0257 - mae: 0.1183 - val_loss: 0.0423 - val_mae: 0.1448
Epoch 39/1000
46/46            0s 5ms/step - loss:
0.0237 - mae: 0.1154 - val_loss: 0.0523 - val_mae: 0.1733
Epoch 40/1000
46/46            0s 5ms/step - loss:
0.0330 - mae: 0.1396 - val_loss: 0.0685 - val_mae: 0.2082
Epoch 41/1000
46/46            0s 4ms/step - loss:
0.0358 - mae: 0.1445 - val_loss: 0.0346 - val_mae: 0.1300
Epoch 42/1000
46/46            0s 5ms/step - loss:
0.0244 - mae: 0.1174 - val_loss: 0.0423 - val_mae: 0.1560
Epoch 43/1000
46/46            0s 5ms/step - loss:
0.0293 - mae: 0.1306 - val_loss: 0.0387 - val_mae: 0.1354
Epoch 44/1000
46/46            0s 4ms/step - loss:
0.0198 - mae: 0.1027 - val_loss: 0.0341 - val_mae: 0.1327
Epoch 45/1000
46/46            0s 5ms/step - loss:
0.0251 - mae: 0.1213 - val_loss: 0.0370 - val_mae: 0.1393
Epoch 46/1000
46/46            0s 4ms/step - loss:
0.0233 - mae: 0.1143 - val_loss: 0.0360 - val_mae: 0.1288
Epoch 47/1000
46/46            0s 5ms/step - loss:
0.0221 - mae: 0.1098 - val_loss: 0.0323 - val_mae: 0.1259
Epoch 48/1000
46/46            0s 5ms/step - loss:
0.0226 - mae: 0.1127 - val_loss: 0.0933 - val_mae: 0.2498
Epoch 49/1000
46/46            0s 4ms/step - loss:
0.0378 - mae: 0.1481 - val_loss: 0.0495 - val_mae: 0.1715
Epoch 50/1000
46/46            0s 4ms/step - loss:
0.0288 - mae: 0.1296 - val_loss: 0.0375 - val_mae: 0.1424
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Raw evaluate results: [0.16016757488250732, 0.3154870867729187]
```

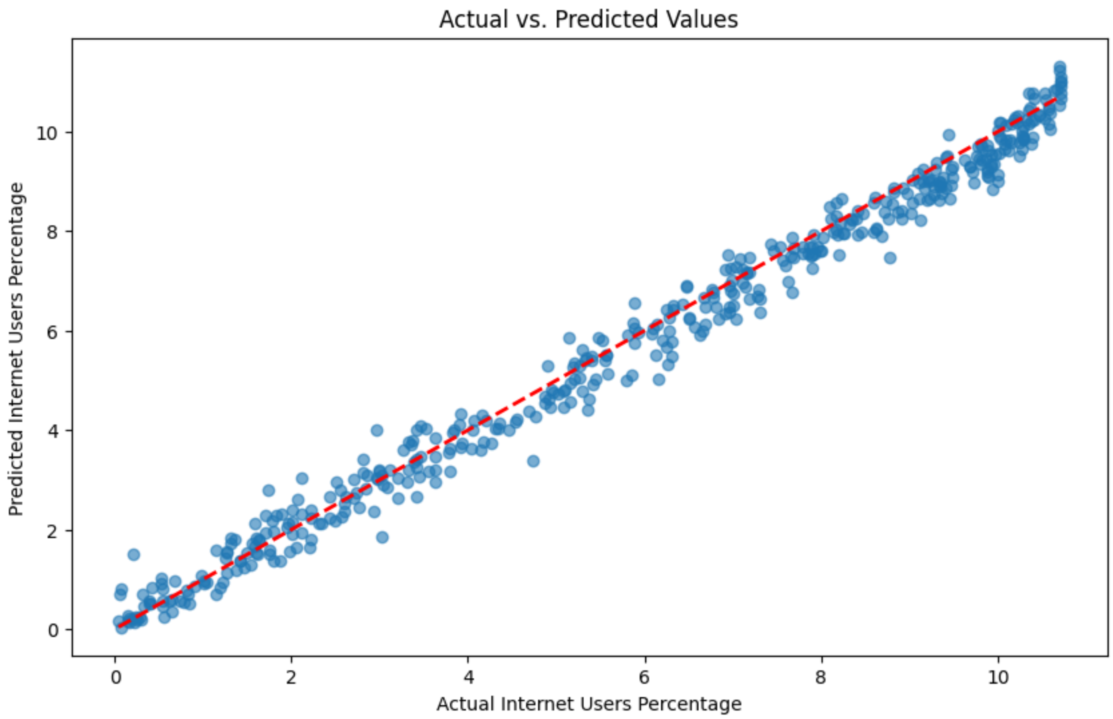


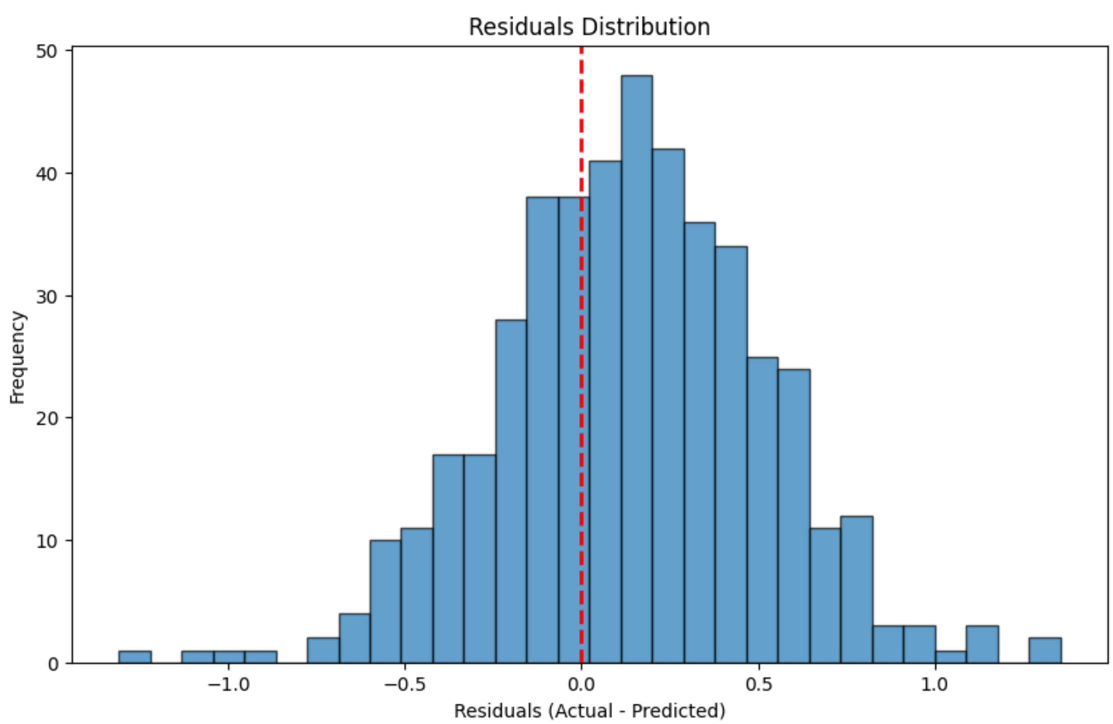
[13]:

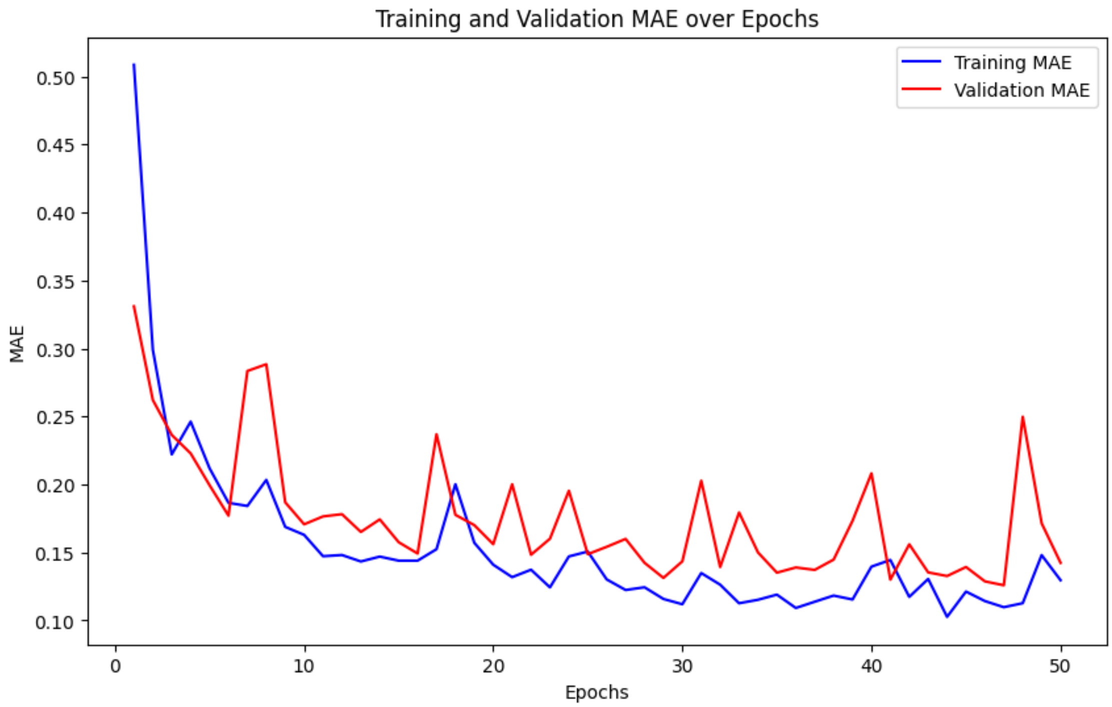
```
# Predictions and Plots
relu_predictions = model_relu.predict(test_features)

if not np.any(np.isnan(relu_predictions)):
    predictions = relu_predictions.flatten()
    final_relu_results = pd.DataFrame({'Actual': test_target, 'Predicted':␣
 ↪predictions})

    # Scatter plot: Actual vs Predicted
    plt.figure(figsize=(10, 6))
    plt.scatter(final_relu_results['Actual'],final_relu_results['Predicted'],␣
 ↪alpha=0.6)
    plt.plot([min(final_relu_results['Actual']),␣
 ↪max(final_relu_results['Actual'])],
             [min(final_relu_results['Actual']),␣
 ↪max(final_relu_results['Actual'])],
             color='red', linestyle='--', lw=2)
    plt.title('Actual vs. Predicted Values')
    plt.xlabel('Actual Internet Users Percentage')
    plt.ylabel('Predicted Internet Users Percentage')
    plt.show()

    # Residual plot
```

```
    final_relu_results['Residuals'] = final_relu_results['Actual'] - final_relu_results['Predicted']
    plt.figure(figsize=(10, 6))
    plt.hist(final_relu_results['Residuals'], bins=30, edgecolor='black', alpha=0.7)
    plt.axvline(x=0, color='red', linestyle='--', lw=2)
    plt.title('Residuals Distribution')
    plt.xlabel('Residuals (Actual - Predicted)')
    plt.ylabel('Frequency')
    plt.show()

    # Training & validation MAE
    mae_relu = history_relu.history['mae']
    val_mae__relu_history = history_relu.history['val_mae']
    epochs_range = range(1, len(mae_sig) + 1)

    plt.figure(figsize=(10, 6))
    plt.plot(epochs_range, mae_relu, 'b', label='Training MAE')
    plt.plot(epochs_range, val_mae__relu_history, 'r', label='Validation MAE')
    plt.title('Training and Validation MAE over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('MAE')
    plt.legend()
    plt.show()

else:
    print("Predictions contain invalid values (NaN or infinity). Cannot plot.")
```

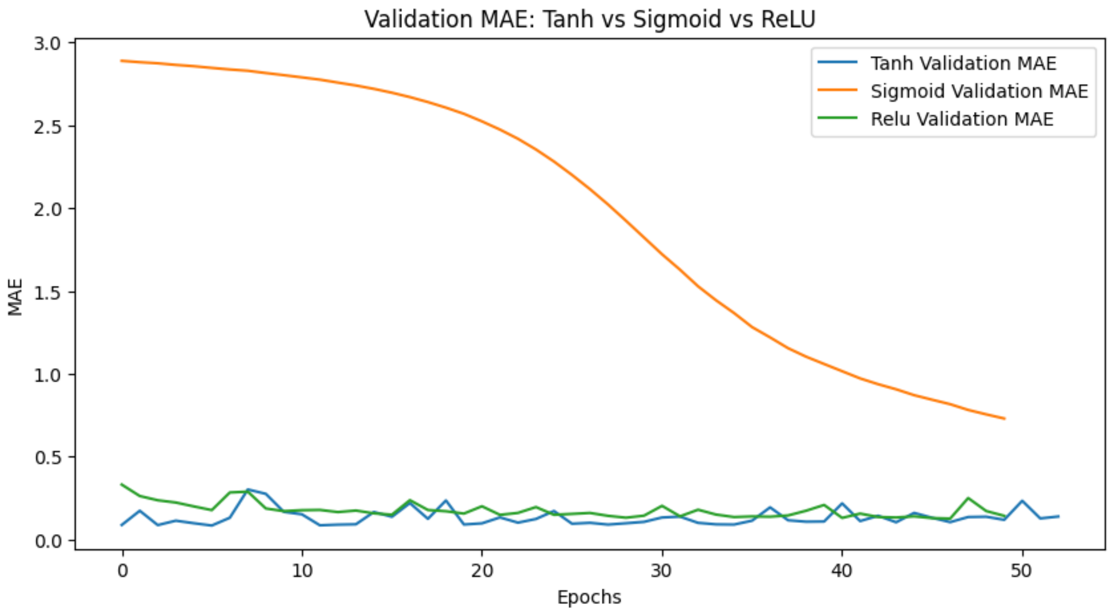
```
15/15                 0s 3ms/step
```


Actual vs. Predicted Values
Predicted Internet Users Percentage
Actual Internet Users Percentage
0
2
4
6
8
10


Residuals Distribution
Frequency
Residuals (Actual - Predicted)
0
10
20
30
40
50
−1.0
−0.5
0.0
0.5
1.0



[14]:
```
plt.figure(figsize=(10,5))
plt.plot(history_tanh.history['val_mae'], label="Tanh Validation MAE")
plt.plot(history_sig.history['val_mae'], label="Sigmoid Validation MAE")
plt.plot(history_relu.history['val_mae'], label="Relu Validation MAE")
plt.title("Validation MAE: Tanh vs Sigmoid vs ReLU")
plt.xlabel("Epochs")
plt.ylabel("MAE")
plt.legend()
plt.show()
```


Validation MAE: Tanh vs Sigmoid vs ReLU
Tanh Validation MAE
Sigmoid Validation MAE
Relu Validation MAE
MAE
Epochs
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
50